PERIODIC PROBABILITY DENSITIES FOR THE FULL VOLTAGE–CONDUCTANCE EQUATION

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Abstract. We construct a nonnegative periodic probability density with nonconstant firing rate for the full voltage–conductance neuron equation, for suitable positive physical parameters. The model retains voltage transport, conductance diffusion, threshold reset, reflection at zero conductance, and firing feedback in both drift and diffusion. A small-noise return-operator comparison and a corrected phase mode give a genuine simple crossing of the full firing response. Analytic parameter selection removes all other integer harmonic resonances, and two free physical parameters close a fixed-period rate equation. A positive prescribed period map then identifies the analytic response with a nonnegative full density. The solution is continuous in time into L^1, has the complete kinetic and conormal boundary traces, finite entropy, and a uniformly finite second conductance moment.

1. Introduction

The voltage–conductance equation describes a population of neurons by a joint probability density in voltage and synaptic conductance. Conductance diffusion supplies the noise, voltage transport carries neurons to threshold, and a boundary reset returns a firing neuron to its resting voltage. The firing flux also changes the conductance drift and diffusion. The resulting autonomous equation can therefore couple fluctuations of the population to its own firing rhythm.

Let $v_E > v_F > 0$ and $y_L, y_*, a_*, c > 0$. On $D = (0, v_F) \times (0, \infty)$ put

$$J(v,y) = y(v_E - v) - y_L v, \qquad y_F = \frac{y_L v_F}{v_E - v_F},$$
$$K_F(t,y) = y_* + cN(t) - y, \qquad a_F(t) = a_* + c^2 N(t). \tag{1.1}$$

The firing rate is $N(t) = \int_{y_F}^{\infty} J(v_F, y) F(t, v_F, y)\,\mathrm{d}y$. We seek a nonnegative mass-one density satisfying

$$\partial_t F + \partial_v(JF) + \partial_y(K_F F) - a_F \partial_{yy} F = 0. \tag{1.2}$$

Writing $J_0(y) = J(0,y)$ and $J_F(y) = J(v_F, y)$, its voltage traces obey

$$\begin{cases} F(t,0,y) = F(t,v_F,y) = 0, & 0 < y < y_F, \\ J_0(y) F(t,0,y) = J_F(y) F(t,v_F,y), & y > y_F. \end{cases} \tag{1.3}$$

The conductance flux $Q = K_F F - a_F F_y$ has zero trace at $y = 0$ and vanishes at infinity. Voltage conditions are kinetic weak traces; the floor condition is the natural conormal condition in the distributional formulation. At infinity we mean that Q paired against every smooth periodic time test and smooth closed-voltage test has tail limit zero. There is no prescribed value at the single grazing conductance y_F.

For this full equation, Fonte Sanchez and Mischler developed the prescribed linear evolution and stationary theory, including exponential relaxation in the frozen linear problem [4, Theorems 1.2 and 2.2]. Fonte Sanchez, Mischler and Salort proved global weak existence for general connectivity [5, Theorem 1.1 and Definition 4.1]. These results give the evolution and trace framework used here. A

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nonconstant periodic trajectory is a separate long-time question: the voltage–conductance discussion in the survey of Carrillo and Roux explicitly lists periodic-solution existence as open [2, Section 7.6]. Carrillo, Dou and Zhou studied a separable simplified equation and its fast-conductance limit; their periodic solutions concern that limit [1, Section 4]. The result below concerns the full nonseparable transport field, positive conductance diffusion, and both firing-feedback terms in (1.1).

Theorem 1.1. *There are positive constants y_*, a_*, c, T such that, with $v_E = 2$, $v_F = 1$ and $y_L = 1$, equations (1.1)–(1.3) admit a nonnegative T-periodic distributional density F of mass one whose firing rate N is nonconstant. The density is continuous in time into $L^1(D)$, has the stated voltage and zero conductance-flux traces, and satisfies*

$$\sup_{0\le t\le T}\int_D \bigl(y^2F(t,v,y) + |F(t,v,y)\log F(t,v,y)|\bigr)\,\mathrm{d}v\,\mathrm{d}y < \infty. \tag{1.4}$$

One may choose $F \in H^1_{\mathrm{per}}(0,T;L^2(D,e^{4y}\,\mathrm{d}v\,\mathrm{d}y))$ and $N \in H^1_{\mathrm{per}}(0,T)$, with nonzero first Fourier coefficient of N. The convention is $0\log 0 = 0$.

The construction uses small frozen conductance diffusion and a mean strictly above the deterministic firing threshold. One firing cycle then provides a useful phase coordinate. We compare its noisy return operator with an OU evolution over the deterministic cycle in a common Gaussian weighted boundary space. The comparison holds in operator norm for arbitrary data in that space. A corrected phase mode, transferred through its isolated one-dimensional projection, gives an actual simple characteristic root. Its real part has a negative noise contribution and a positive mean-feedback contribution. The reflecting-floor part of diffusion feedback is retained throughout.

Two further points are needed to obtain the theorem. First, the fixed-noise physical resolvent is analytic in the mean and diffusion, so a countable zero-set argument selects parameters with no other integer harmonic resonance. The mean and coupling then serve as two free parameters in a fixed-period rate equation. Second, its initially signed density response is identified with the fixed point of a positive prescribed period map. This step establishes nonnegativity without requiring a pointwise positive lower bound for the stationary density. Stability of the resulting periodic solutions and explicit numerical parameter values are not determined by this construction.

Section 2 constructs the divergence inverse with its complete outgoing trace and natural floor input. Section 3 establishes the process and cycle estimates. Section 4 proves the simple full-response crossing, and Section 5 closes the periodic rate equation and proves the theorem. Appendix A gives the boundary and process constructions; Appendix B supplies the bounded-clock comparison.

All complex transpose pairings below are bilinear; only energy inner products use conjugation. We keep separate the random first hit τ_{f}, deterministic cycle time τ, and final period T. Constants in fixed-positive-noise resolvent estimates may depend on that noise. Small-noise estimates are uniform only on the specified compact mean and frequency sets, and harmonic expansions are for each fixed harmonic.

2. The linear equation and the firing trace

For the construction we fix

$$v_E = 2, \qquad v_F = 1, \qquad y_L = 1. \tag{2.1}$$

Thus $D = (0,1)\times(0,\infty)$, $J(v,y) = y(2-v) - v$, $y_F = 1$, $J_0(y) = 2y$ and $J_F(y) = y - 1$. We first keep the conductance mean and diffusion prescribed. Write them as $\mu(t)$ and $\mathfrak{a}(t)$, bounded above and below by positive constants. The conductance flux is

$$Q_{\mu,\mathfrak{a}} f = (\mu - y)f - \mathfrak{a} f_y.$$

The weak formulation we use retains this flux at the reflecting boundary. If $\gamma_0 f, \gamma_F f$ are the voltage traces, they satisfy

$$\gamma_0 f = \gamma_F f = 0 \quad (0 < y < 1), \qquad J_0\gamma_0 f = J_F\gamma_F f \quad (y > 1). \tag{2.2}$$

For a smooth test φ, compactly supported in y and allowed to be nonzero at all physical faces, the equation reads

$$\begin{aligned}\left[\int_D f\varphi\right]_{t_1}^{t_2} + \int_{t_1}^{t_2}\int_1^\infty J_F\gamma_F f\,[\varphi(t,1,y) - \varphi(t,0,y)]\,\mathrm{d}y\,\mathrm{d}t \\ = \int_{t_1}^{t_2}\int_D f\,[\varphi_t + J\varphi_v + (\mu - y)\varphi_y] - \mathfrak{a} f_y\varphi_y.\end{aligned} \tag{2.3}$$

There is no boundary term at $y = 0$. This is the zero conormal-flux condition. For periodic solutions the endpoint term cancels when the test is periodic. All voltage conditions are understood almost everywhere in their kinetic trace measures. No trace value is needed at $y = 1$.

2.1. **The frozen evolution.** Let m belong to $I = [2, 3]$ and let $a > 0$. Fix $\sigma > 1$ with $2a\sigma < m$; we eventually take $\sigma = 2$. Define

$$W(y) = e^{2\sigma y}, \qquad \mathsf{H}_\sigma = L^2(D, W\,\mathrm{d}v\,\mathrm{d}y), \qquad \mathsf{H}_{-\sigma} = L^2(D, W^{-1}\,\mathrm{d}v\,\mathrm{d}y).$$

$$\mathsf{H}_{\sigma,0} = \left\{h \in \mathsf{H}_\sigma : \int_D h = 0\right\}.$$

We complexify these spaces. Their transpose pairing is the bilinear integral $\langle h, u\rangle = \int_D hu$; conjugation is used only in Hilbert energy identities. The mass functional is bounded on H_σ.

We use the following consequences of the prescribed linear theory of Fonte Sanchez and Mischler.

Proposition 2.1 (The frozen linear evolution)**.** *For prescribed positive bounded $\mu, \mathfrak{a}$ and initial data in H_σ, there is a unique solution of the linear equation in $C([0, t_*]; \mathsf{H}_\sigma)$, with local L^2 conductance derivative and the renormalized weak traces, for every finite t_*. It preserves positivity and mass. For $\mu = m$, $\mathfrak{a} = a$, its evolution is a strongly continuous semigroup $S(t)$ on H_σ with generator $\mathcal{L} = \mathcal{L}_{m,a}$. There is a normalized nonnegative invariant density $M = M_{m,a}$, belonging to every fixed exponential weighted space, and constants $C_a, \kappa_a > 0$ such that*

$$\|S(t)h\|_{\mathsf{H}_\sigma} \le C_a e^{-\kappa_a t}\,\|h\|_{\mathsf{H}_\sigma}, \qquad h \in \mathsf{H}_{\sigma,0}. \tag{2.4}$$

The stationary density is in $D(\mathcal{L})$ and $\mathcal{L}M = 0$.

Proof. The prescribed statement and its renormalized Green identity are [4, Theorem 2.2 and (2.18)–(2.19)]. The stationary and decay statements follow from [4, Theorem 1.2] by setting the source's connectivity to zero and its background mean and diffusion to m, a. In particular its coefficients are then exactly $m - y, a$. The invariant state is the generator-domain stationary state constructed in Section 7 of that paper. Autonomy and uniqueness give the semigroup property; the asserted time continuity gives strong continuity. Its $p = 2$ decay estimate is (2.4). These are estimates for the frozen equation. The firing feedback will be included in Section 4. □

The domain $D(\mathcal{L})$ means the strong semigroup generator domain. We do not replace it by a prescribed collection of smooth functions at the two characteristic corners. The differential expressions in the interior are

$$\mathcal{L}h = -\partial_v(Jh) - \partial_y((m - y)h) + ah_{yy}, \qquad \mathcal{G}u = Ju_v + (m - y)u_y + au_{yy}.$$

The outgoing space and the firing functional are

$$\mathsf{O}_\sigma = L^2((1, \infty), J_F W\,\mathrm{d}y), \qquad \ell h = \int_1^\infty J_F(y)h_F(y)\,\mathrm{d}y. \tag{2.5}$$

The next estimate defines these quantities on the actual generator domain.

Lemma 2.2 (Forward energy and the generator trace)**.** *There is a constant $C = C_{m,a,\sigma}$ such that*

$$\|S(t)h\|_{\mathsf{H}_\sigma}^2 + 2a\int_0^t \|\partial_y S(s)h\|_{\mathsf{H}_\sigma}^2 \,\mathrm{d}s + \int_0^t\int_1^\infty J_F(1-J_F/J_0)|(S(s)h)_F|^2 W \,\mathrm{d}y\,\mathrm{d}s$$
$$\le \|h\|_{\mathsf{H}_\sigma}^2 + C\int_0^t \|S(s)h\|_{\mathsf{H}_\sigma}^2 \,\mathrm{d}s. \tag{2.6}$$

For $h \in D(\mathcal{L})$, its derivative and outgoing trace satisfy

$$2a\,\|h_y\|_{\mathsf{H}_\sigma}^2 + \tfrac{1}{2}\,\|h_F\|_{\mathsf{O}_\sigma}^2 \le C\,\|h\|_{\mathsf{H}_\sigma}^2 - 2\,\mathrm{Re}(\mathcal{L}h, h)_{\mathsf{H}_\sigma}. \tag{2.7}$$

In particular ℓ is continuous for the graph norm of $D(\mathcal{L})$. The finite-time version of (2.6) *is uniform for prescribed coefficients in a sufficiently small positive neighborhood of m, a.*

Proof. Here are the truncations in the Green identity. For real f use

$$b_n(s) = 2n^2\big(\sqrt{1+s^2/n^2}-1\big), \qquad D_n(s) = sb_n'(s) - b_n(s).$$

These are smooth convex functions with bounded first and second derivatives, $0 \le D_n \le b_n \le s^2$, and $b_n'' \to 2$. If boundedness of the renormalization itself is required, first compose b_n with a smooth bounded argument truncation at height A. For fixed n its first and second derivatives are bounded independently of A. On compact physical intervals the local L^2 derivative and the source trace bounds dominate every term, so $A \to \infty$ recovers b_n. This also applies to the entropy renormalizations used later.

Use b_n with W cut off between R and $2R$. At high conductance the reset condition, with $r = J_F/J_0 \in (0,1)$, makes the boundary loss

$$J_F\big[b_n(f_F) - r^{-1}b_n(rf_F)\big] \ge 0.$$

It converges to $J_F(1-r)|f_F|^2$. Integrate the cross diffusion term as $-a\int \partial_y b_n(f)W_y$. At the floor this gives $-mD_n(f_0) + 2a\sigma b_n(f_0)$. The source local H_y^1 class gives $f_0 \in L^2(\,\mathrm{d}t\,\mathrm{d}v)$, so this term converges to $-(m-2a\sigma)|f_0|^2$. Its sign is used after, rather than before, the truncation is removed.

Away from the cutoff the volume term is

$$(y+2)D_n(f) + 2\sigma(m-y)b_n(f) + 4a\sigma^2 b_n(f) \le C|f|^2,$$

since $2\sigma > 1$. The y-cutoff errors are bounded by a weighted H_σ tail: on their support y/R is bounded. Pass $n \to \infty$ for fixed R, then $R \to \infty$. Fatou's lemma retains the nonnegative gradient and reset losses. This proves (2.6) and also the favorable floor loss. Apply the argument to real and imaginary parts.

For $h \in D(\mathcal{L})$ divide by t. The norm difference quotient tends to $2\,\mathrm{Re}(\mathcal{L}h, h)$. The averages $t^{-1}\int_0^t S(s)h\,\mathrm{d}s$ tend to h in graph norm. Jensen's inequality and the preceding estimate bound their gradients and traces weakly; their averaged Green identities identify the limits. As $1 - J_F/J_0 \ge 1/2$, weak lower semicontinuity gives (2.7). Finally

$$|\ell h| \le \left(\int_1^\infty J_F W^{-1}\,\mathrm{d}y\right)^{1/2} \|h_F\|_{\mathsf{O}_\sigma}.$$

All coefficients and the limiting floor sign remain controlled in a small positive prescribed-coefficient neighborhood. □

2.2. **Divergence inputs and the reflecting boundary.** The linearized feedback is a conductance divergence. For compact smooth g supported inside D, put $Bg = \partial_y g$. The completion for general $g \in \mathsf{H}_\sigma$ will be defined by

$$\langle Bg, \varphi\rangle = -\int_D g\varphi_y, \tag{2.8}$$

including tests nonzero at the floor. In particular it does not assign a point value to $g(0)$.

By (2.4), on the zero-mass space the ordinary inverse is

$$R_{i\theta}h = \int_0^\infty e^{-i\theta t} S(t)h \,\mathrm{d}t, \qquad \theta \in \mathbb{R}. \tag{2.9}$$

It maps into $D(\mathcal{L})$ and $(i\theta - \mathcal{L})R_{i\theta}h = h$. This follows by taking the difference quotient of the integral under $S(s)$; the omitted integral over $(0, s)$ tends to h after division by s. The gap also excludes a homogeneous zero-mass solution.

We first bound divergence inputs independently of a killed backward harmonic. The bounded-source backward construction needed in this proof is given in Appendix A; in particular it proves the actual Neumann condition before taking a global dual energy estimate.

Proposition 2.3 (Completed divergence resolvent)**.** *There is a bounded map*

$$\mathcal{R}_\theta : g \longmapsto (h, h_y, h_F) \quad \textit{from } \mathsf{H}_\sigma \textit{ to } \mathsf{E}_\sigma := \mathsf{H}_{\sigma,0} \times \mathsf{H}_\sigma \times \mathsf{O}_\sigma,$$

uniformly for all $\theta \in \mathbb{R}$, whose density component extends $h = R_{i\theta}Bg$ for compact smooth g. Its weak equation is

$$\begin{aligned} i\theta \int_D h\varphi - \int_D h[J\varphi_v + (m-y)\varphi_y] + a\int_D h_y\varphi_y + \int_1^\infty J_F h_F(\varphi_F - \varphi_0)\,\mathrm{d}y \\ = -\int_D g\varphi_y. \end{aligned} \tag{2.10}$$

Thus the natural flux $(m-y)h - ah_y - g$ has zero floor trace. The original voltage conditions hold in the completed trace sense. We continue to denote the density component by $R_{i\theta}Bg$. The maps extend holomorphically to a neighborhood of each fixed imaginary frequency where the ordinary zero-mass resolvent exists. Constants and neighborhood sizes may depend on the fixed $a > 0$.

Proof. Let s exceed the full semigroup growth bound and a fixed dual-energy constant. Here $Z_t = (V_t, Y_t)$ has reflected conductance $\mathrm{d}Y_t = (m - Y_t)\,\mathrm{d}t + \sqrt{2a}\,\mathrm{d}W_t + \mathrm{d}K_t$ and voltage $\dot{V}_t = J(V_t, Y_t)$; a threshold hit resets voltage from 1 to 0 without changing conductance. Proposition 3.1 establishes its density law, and Proposition A.1 supplies the transpose-resolvent representation below. For bounded compact smooth r the full-process value

$$u(z) = \mathbb{E}_z \int_0^\infty e^{-(s+i\theta)t} r(Z_t)\,\mathrm{d}t$$

is the transpose of the full forward resolvent. It is bounded by $\|r\|_\infty / s$. Proposition A.1 in the appendix gives, by localization at the corners and the actual Neumann floor,

$$(s - C)\,\|u\|^2_{\mathsf{H}_{-\sigma}} + a\,\|u_y\|^2_{\mathsf{H}_{-\sigma}} \le \|r\|_{\mathsf{H}_{-\sigma}}\,\|u\|_{\mathsf{H}_{-\sigma}}\,. \tag{2.11}$$

Completion of compact smooth r, using closedness of the distributional derivative, gives the same estimate for every $r \in \mathsf{H}_{-\sigma}$.

For an arbitrary such r set $r_0 = r - (\int_D Mr)\mathbf{1}$. The transpose of (2.9), represented in $\int_D M\psi = 0$, gives

$$(i\theta - \mathcal{L}^T)\psi = r_0, \qquad \|\psi\|_{\mathsf{H}_{-\sigma}} \le C_a\,\|r\|_{\mathsf{H}_{-\sigma}}\,.$$

The representative is an actual transpose-domain element: decompose $h \in D(\mathcal{L})$ into $(\int h)M + h_0$. The stationary terms vanish by the two mean-zero normalizations. The shifted identity is

$$\psi = (R^{\mathrm{full}}_{s+i\theta})^T(r_0 + s\psi).$$

Its input is bounded uniformly in θ. Applying (2.11) at this same frequency proves $\|\psi_y\|_{\mathsf{H}_{-\sigma}} \le C_a\,\|r\|_{\mathsf{H}_{-\sigma}}$.

For compact g, $h = R_{i\theta}Bg \in D(\mathcal{L})$ and transpose duality now gives

$$\langle h, r\rangle = \langle Bg, \psi\rangle = -\int_D g\psi_y.$$

Taking the dual norm bounds $\|h\|_{\mathsf{H}_\sigma}$ by $C_a \|g\|_{\mathsf{H}_\sigma}$. Use (2.7) with $\mathcal{L}h = i\theta h - Bg$. The remaining source term is

$$(Bg, h)_{\mathsf{H}_\sigma} = -\int_D g(\overline{h_y} + 2\sigma\overline{h})W.$$

Young's inequality absorbs its derivative term. Hence

$$\|h\|_{\mathsf{H}_\sigma} + \|h_y\|_{\mathsf{H}_\sigma} + \|h_F\|_{\mathsf{O}_\sigma} \le C_a \|g\|_{\mathsf{H}_\sigma}, \tag{2.12}$$

uniformly in θ, including zero.

Approximate general g by compact smooth functions away from $y = 0$. Apply (2.12) to differences. All three components converge strongly, and their weak equations give (2.10). This is why the inhomogeneous floor flux is retained even though the approximants vanish there. The product E_σ is an ambient Hilbert space; derivative and trace identities are consequences of the weak equation for these outputs.

Finally, for $\lambda_0 = i\theta$ use

$$R_\lambda B = R_{\lambda_0} B + (\lambda_0 - \lambda) R_\lambda R_{\lambda_0} B.$$

The second term has an ordinary $\mathsf{H}_{\sigma,0}$ datum, so the ordinary resolvent identity and domain energy give the local holomorphic extension. □

We will also need a finite-time form of the same completion.

Lemma 2.4 (Causal divergence forcing)**.** *For $g \in L^2((0,t_*);\mathsf{H}_\sigma)$ there is a unique zero-initial-data solution of $(\partial_t - \mathcal{L})w = Bg$ in the completed energy class, with*

$$\sup_{0\le t\le t_*} \|w(t)\|^2_{\mathsf{H}_\sigma} + \int_0^{t_*} \left(\|w_y\|^2_{\mathsf{H}_\sigma} + \|w_F\|^2_{\mathsf{O}_\sigma}\right) \mathrm{d}t \le C_{a,m,t_*} \|g\|^2_{L^2_t \mathsf{H}_\sigma}. \tag{2.13}$$

Its mass is zero and its natural base flux equals g at the floor in the weak formulation.

Proof. First take g continuously differentiable in time and compactly supported inside D. Its divergence is a differentiable H_σ forcing, so the semigroup convolution is a classical $D(\mathcal{L})$ solution. This also follows by integrating the time derivative of Bg in that convolution. Apply (2.7) to $w(t)$. Integrating the input as above, Young's inequality and Gronwall give (2.13). Complete the inputs in $L^2_t\mathsf{H}_\sigma$. The outputs converge in $C_t\mathsf{H}_\sigma$, and their derivatives and traces converge in L^2. Their weak equations retain the same floor input. The difference of two completions is a homogeneous zero-initial-data solution, so prescribed uniqueness in Proposition 2.1 proves uniqueness. All these solutions lie in its local energy class; a smaller exponential weight may be used for this uniqueness comparison. □

3. The reset process and one firing cycle

The first-return description will be used to locate a characteristic root. We therefore identify both the state law and the firing-count measure, rather than using a formal stochastic representation.

3.1. The process, the first hit, and the outgoing law.

Let $Z_t = (V_t, Y_t)$, where

$$\mathrm{d}Y_t = (m - Y_t)\,\mathrm{d}t + \sqrt{2a}\,\mathrm{d}W_t + \mathrm{d}K_t, \qquad Y_t \ge 0, \qquad \dot{V}_t = J(V_t, Y_t). \tag{3.1}$$

Here K is the minimal continuous regulator at zero. Whenever V reaches 1 with $Y > 1$, it is reset to zero without changing Y. Write t_j for these successive hits and τ_{f} for the next hit, including when the initial voltage is not zero. The half-line Skorokhod map constructs Y pathwise; the voltage equation then constructs the path between hits. The absence of grazing and accumulation of hits is included in the next proposition.

Proposition 3.1 (Law and firing counts)**.** *For a nonnegative $h \in \mathsf{H}_\sigma$ of mass one, the process* (3.1) *has density $S(t)h$ at every deterministic time. Almost surely every threshold hit is outgoing and has $Y_{t_j} > 1$. There are finitely many hits on each bounded time interval, with finite expected count. For every bounded Borel function q,*

$$\mathbb{E}_h \sum_{t_j \le t} q(t_j, Y_{t_j}) = \int_0^t \int_1^\infty q(s,y) J_F(y) (S(s)h)_F(y) \,\mathrm{d}y \,\mathrm{d}s. \tag{3.2}$$

The same identity applies to integrable unbounded functions by exhaustion.

The proof in Appendix A lifts the forward density to a circle times a signed conductance line, applies Euclidean superposition, and folds the conductance. A conditional-density argument excludes grazing before identifying the circle path with the reset path. Finally the voltage test $vq(t,y)$ identifies the entire count measure. This proves the statement for the prescribed semigroup class in Proposition 2.1; no converse for arbitrary weak solutions is needed.

We now restrict m to $I = [2,3]$ and decrease $a_0 > 0$ as necessary. Set

$$\varepsilon = \sqrt{a}, \qquad \eta = \frac{y-m}{\varepsilon}, \qquad \mathcal{A} = \partial_{\eta\eta} - \eta\partial_\eta, \qquad w_\beta(\eta) = e^{\beta\eta^2}.$$

The reflecting floor is $\eta = -m/\varepsilon$.

Lemma 3.2 (Uniform first-hit estimates)**.** *There is $r_* > 0$ such that*

$$\sup_{\substack{m\in I,\, 0<a\le a_0 \\ z\in[0,1)\times[0,\infty)}} \mathbb{E}_z e^{r_* \tau_{\mathrm{f}}} < \infty. \tag{3.3}$$

For $0 < \beta < 1/2$, after reducing r_* if needed, every stopping time $S \le \tau_{\mathrm{f}}$ satisfies

$$\mathbb{E}_z e^{r_* S} w_\beta(\eta_S) \le C_\beta w_\beta(\eta_0), \tag{3.4}$$

$$\mathbb{E}_z \int_0^{\tau_{\mathrm{f}}} e^{r_* s} w_\beta(\eta_s) \,\mathrm{d}s \le C_\beta w_\beta(\eta_0). \tag{3.5}$$

Constants are uniform on this small-noise family.

Proof. Choose $\delta > 0$ with $4\delta < \min_I(m-1)$ and choose t_0 such that $m(1-e^{-t_0}) \ge 1 + 2\delta$ on I. Couple Y with the unreflected OU process U starting from zero. The explicit equations give

$$Y_t - U_t = e^{-t} Y_0 + \int_0^t e^{-(t-s)} \,\mathrm{d}K_s \ge 0, \qquad U_t = m(1-e^{-t}) + \varepsilon\zeta_t.$$

For $B = t_0 + 1/\delta$, the stochastic-integral isometry and the maximal inequality imply $\mathbb{E}\sup_{[0,B]} |\zeta_t|^2 \le C_B$. For small a, with probability at least $1/2$, $Y \ge 1+\delta$ throughout $[t_0, B]$. On this event, before a hit, $J(V,Y) \ge J(1, 1+\delta) = \delta$. A hit must occur by B, from every initial state. Independence of the Brownian increments on successive blocks gives $\mathbb{P}_z(\tau_{\mathrm{f}} > nB) \le 2^{-n}$ and hence (3.3).

For the weights,

$$\mathcal{A} w_\beta = [2\beta - 2\beta(1-2\beta)\eta^2] w_\beta. \tag{3.6}$$

The regulator term is nonpositive because w_β' is negative at the floor. The expression $(\mathcal{A} + r_*) w_\beta$ is bounded above uniformly in the floor location. Stop at a finite time, a finite upper conductance and S. Itô's formula and (3.3) give (3.4) by Fatou's lemma. The stronger inequality $(\mathcal{A} + r_*) w_\beta \le C_\beta - w_\beta$, which follows from the same negative quadratic term, gives (3.5). □

Let $N_0 = \ell M$ be the stationary rate, and define the outgoing mark law

$$\pi_a(\,\mathrm{d}y) = N_0^{-1} J_F(y) M_F(y) \,\mathrm{d}y, \qquad y > 1. \tag{3.7}$$

We also write π_a for its image under $y \mapsto (y-m)/\sqrt{a}$. This law describes conductance at a firing time, whereas the conductance marginal of M describes a typical time.

Lemma 3.3 (Stationary moments and cycle occupation)**.** *The conductance marginal is exactly*

$$p_a(y) = C_a \exp[-(y-m)^2/(2a)]\, \mathbf{1}_{\{y\geq 0\}}. \tag{3.8}$$

For $L = \log 2$, the stationary rate and mark law satisfy

$$\frac{m-1}{L} \leq N_0 \leq \frac{\mathbb{E}_M Y}{L}, \qquad \int w_\beta \,\mathrm{d}\pi_a \leq C_\beta \quad (0<\beta<1/2). \tag{3.9}$$

For every nonnegative measurable g, or every integrable complex g,

$$\int_D Mg = N_0\, \mathbb{E}_{\pi_a} \int_0^{\tau_{\mathrm{f}}} g(Z_s)\,\mathrm{d}s. \tag{3.10}$$

The cycle starts at voltage zero with conductance distributed as π_a. In particular $N_0 \mathbb{E}_{\pi_a} \tau_{\mathrm{f}} = 1$.

Proof. Integrating the stationary equation in voltage cancels the reset flux. The marginal has zero current, so $ap_a' = (m-y)p_a$ and normalization gives (3.8). In the η variable it is a standard Gaussian restricted to $\eta \geq -m/\varepsilon$, with normalization at least $1/2$. Thus

$$\int_D M(1+|\eta|^j) w_\beta \leq C_{\beta,j}, \qquad \mathbb{E}_M Y \geq m.$$

Put $X = \log[2/(2-v)]$ and $b(X) = e^X - 1$. Between hits, $\dot X = Y - b(X)$; a reset decreases X by L. The stationary identity with this bounded voltage test is $LN_0 = \mathbb{E}_M[Y - b(X)]$. Since $0 \leq b \leq 1$, it gives the first bounds in (3.9).

To control the different measure π_a, use the concave truncations $\zeta_n(s) = n(1-e^{-s/n})$ and test with $(X/L)\zeta_n(w_\beta)$. The reset loss is $\zeta_n(w_\beta)$. Its floor derivative is nonpositive, so the regulator has the favorable sign. Concavity and (3.6) imply

$$\mathcal{G}\big[(X/L)\zeta_n(w_\beta)\big] \leq (Y/L + 2\beta) w_\beta.$$

Stationarity, followed by monotone convergence, bounds $N_0 \pi_a(w_\beta)$ by the preceding Gaussian moment. This proves the second assertion in (3.9). Truncating in y first justifies this test in the forward identity.

For bounded g, put $k_g(z) = \mathbb{E}_z \int_0^{\tau_{\mathrm{f}}} g(Z_s)\,\mathrm{d}s$ and $k_{g,0}(y) = k_g(0,y)$. The first-hit estimate makes k_g bounded. Dynamic programming, patched over the finitely many cycles, gives

$$\mathbb{E}_h[k_g(Z_t) - k_g(Z_0)] = -\mathbb{E}_h \int_0^t g(Z_s)\,\mathrm{d}s + \mathbb{E}_h \sum_{t_j \leq t} k_{g,0}(Y_{t_j}).$$

Start at M and apply (3.2). The left side is zero, and division by t proves (3.10). Monotone convergence [7, Theorem 1.4.44] and then domination give the asserted extensions. □

3.2. **Killed values and the generator pairing.** Fix henceforth $\beta = 1/32$. The boundary space is

$$\mathcal{B}_\beta = \{f \in C(\mathbb{R};\mathbb{C}) : f(\eta)/w_\beta(\eta) \longrightarrow 0 \text{ as } |\eta| \to \infty\}, \qquad \|f\|_\beta = \sup_\eta |f(\eta)|/w_\beta(\eta).$$

Choose $r_0 > 0$ so that $4r_0 < r_*$. For $|\operatorname{Re}\lambda| \leq r_0$, define

$$V_\lambda f(v,y) = \mathbb{E}_{v,y}[e^{-\lambda\tau_{\mathrm{f}}} f(\eta_{\tau_{\mathrm{f}}})], \qquad P_a(\lambda) f(\eta) = V_\lambda f(0, m+\varepsilon\eta). \tag{3.11}$$

For $\eta < -m/\varepsilon$ the last expression means its value at the floor. Estimates (3.4)–(3.5) also apply when an interior reward is added. In particular a value with terminal bound Fw_β and source bound Rw_β is bounded by $C(F+R)w_\beta$. Discount derivatives follow from $t^j e^{r_0 t} \leq j!(r_* - r_0)^{-j} e^{r_* t}$, giving local norm holomorphy. Continuity of point-start values and the local backward equation are proved in Appendix A.

The invariance of π_a under $P_a(0)$ follows without an additional renewal theorem. For a bounded terminal f, patch the harmonic $V_0 f$ over cycles and start at M. Its mean reset jump is $\pi_a(f - P_a(0)f)$,

hence is zero by (3.2). Monotone convergence extends this to nonnegative data, and domination to π_a-integrable complex data. In particular it applies to $|f|^2$ for $f \in \mathcal{B}_\beta$.

Proposition 3.4 (A constant reset jump)**.** *Suppose $u = V_\lambda f$ has the constant outgoing jump*

$$u(1,y) - u(0,y) = A, \qquad y > 1. \tag{3.12}$$

Then u has at most polynomial growth at infinity, belongs to $\mathsf{H}_{-\sigma}$, and for every $h \in D(\mathcal{L})$,

$$\langle \mathcal{L}h, u\rangle = \lambda\langle h, u\rangle - A\ell h. \tag{3.13}$$

If $A = 0$, this is the actual transpose-domain relation $u \in D(\mathcal{L}^T)$, $\mathcal{L}^T u = \lambda u$. At a frequency where the zero-mass resolvent exists, the independently completed inverse in Proposition 2.3 gives

$$u_y \in \mathsf{H}_{-\sigma}, \qquad A\ell R_\lambda B g = -\int_D g u_y, \quad g \in \mathsf{H}_\sigma. \tag{3.14}$$

Proof. Appendix A proves polynomial growth by stopping at a fixed large conductance and estimating discounted counts before that stop. It also constructs the local Neumann floor. Patch the killed martingales over cycles to obtain

$$u(z) = \mathbb{E}_z[e^{-\lambda t}u(Z_t)] + A\mathbb{E}_z \sum_{t_j \le t} e^{-\lambda t_j}.$$

The required domination follows from polynomial growth, finite moments of the reflected OU path supremum, and the pathwise bound

$$L\,\#\{j : t_j \le t\} \le X_0 + \int_0^t Y_s \,\mathrm{d}s. \tag{3.15}$$

Pair with h and use (3.2), splitting signed data into positive densities when necessary:

$$\langle S(t)h, u\rangle = e^{\lambda t}\langle h, u\rangle - A\int_0^t e^{\lambda(t-s)}\ell S(s)h \,\mathrm{d}s.$$

For $h \in D(\mathcal{L})$, graph continuity of the trace in Lemma 2.2 permits differentiation at zero. This proves (3.13).

For compact smooth g in the interior set $h = R_\lambda B g$. The resolvent equation and (3.13) imply $A\ell h = \int_D u\,\partial_y g$. Proposition 2.3 bounds this functional by $C|A|\,\|g\|_{\mathsf{H}_\sigma}$. Weighted L^2 Riesz representation therefore supplies the distributional derivative $u_y \in \mathsf{H}_{-\sigma}$. Complete the inputs g by the triple estimate (2.12); this gives (3.14), including its natural floor flux. The derivative of a killed harmonic was not used to construct that inverse. □

3.3. **Cycle energies and the feedback functional.** For an interior source we use smooth functions on the closed physical half-strip: on each finite part they admit smooth local extensions, with the two voltage faces treated separately. No matching condition on the source across reset is imposed. If $f \in \mathcal{B}_\beta$ and $|r(v,y)| \le R w_\beta((y-m)/\sqrt{a})$, define the inhomogeneous killed value by

$$u(z) = \mathbb{E}_z\left[e^{-\lambda\tau_{\mathrm{f}}} f(\eta_{\tau_{\mathrm{f}}}) + \int_0^{\tau_{\mathrm{f}}} e^{-\lambda s} r(Z_s)\,\mathrm{d}s\right].$$

The source is assumed smooth as just specified; a weighted size bound alone is not a regularity assumption. The local construction in Appendix A gives $\mathcal{G}u = \lambda u - r$ and the reflecting derivative for these values. This is the source class used in the next lemma and in its forced-feedback consequence.

Lemma 3.5 (Stationary Fisher and killed-value energy)**.** *Uniformly for $m \in I$, $0 < a \le a_0$,*

$$a\int_D \frac{|M_y|^2}{M} \le \mathbb{E}_M Y + 2 \le C. \tag{3.16}$$

The quotient is defined as zero where numerator and denominator vanish. If u is the killed value above with the stated smooth source, terminal norm $F = \|f\|_\beta$ and source bound Rw_β, then

$$\int_D M|u|^2 + a\int_D M|u_y|^2 \le C(F+R)^2, \qquad |\operatorname{Re}\lambda| \le r_0. \tag{3.17}$$

Proof. For the first estimate use the smooth convex entropy caps

$$b_{\delta,n}(0) = b'_{\delta,n}(0) = 0, \qquad b''_{\delta,n}(s) = \frac{1}{(s+\delta)(1+s/n)^2}, \quad s \ge 0.$$

Their even extensions are C^2; their first derivatives are bounded for fixed δ, n, and $0 \le sb'_{\delta,n}(s) - b_{\delta,n}(s) \le s$. The bounded-argument extension in the proof of Lemma 2.2 makes these valid renormalizations. Apply them to the stationary equation, first with a large-y cutoff. The floor loss and the reset loss are nonnegative by convexity. For fixed n, δ the exponential tails of M, M_y remove the cutoff. Drop these losses, let $n \to \infty$ and then $\delta \downarrow 0$. The remaining volume coefficient is $y + 2$, giving (3.16) by Fatou's lemma. Interior smoothness and nonnegativity imply $M_y = 0$ at an interior zero of M; no boundary lower bound is used.

For the second estimate, the local equation and constructed floor permit Itô's formula for $|u|^2$ before $\tau_{\mathfrak{f}}$:

$$\mathcal{G}|u|^2 = 2\operatorname{Re}\lambda\,|u|^2 - 2\operatorname{Re}(r\overline{u}) + 2a|u_y|^2.$$

Stop away from the grazing corners and at bounded conductance and time. Point starts have no grazing hit, and a path cannot reach the floor at voltage zero. The estimates with 4β and $4r_0$ give uniform integrability for the endpoint terms; Fatou retains the gradient. Start the cycle with π_a. Its invariance for $P_a(0)$, followed by (3.10), yields

$$a\int_D M|u_y|^2 \le \frac{N_0}{2}\pi_a|f|^2 + |\operatorname{Re}\lambda|\int_D M|u|^2 + \left|\int_D Mr\overline{u}\right|.$$

The reward bound, the exact Gaussian marginal, and $\pi_a w_{2\beta} \le C$ bound every term on the right by $C(F+R)^2$. They also bound $\int M|u|^2$, proving (3.17). □

For a complex coupling c define the complete weak feedback by

$$\mathfrak{b}_{\lambda,c}(f) = c\int_D M(V_\lambda f)_y - c^2\int_D M_y(V_\lambda f)_y. \tag{3.18}$$

Its second term represents diffusion feedback, including the reflecting floor. It is defined by first derivatives. For a smooth Neumann test u with integrable tails it equals $c^2\int Mu_{yy}$.

Corollary 3.6 (Small feedback and holomorphy)**.** *On the preceding strip,*

$$\|\mathfrak{b}_{\lambda,c}\|_{\mathcal{B}^*_\beta} \le C\left(\frac{|c|}{\sqrt{a}} + \frac{|c|^2}{a}\right). \tag{3.19}$$

For a terminal-zero value with a smooth source in the same class and source norm R, its feedback is bounded by the same right side times R. The functional is norm holomorphic in λ and polynomial in c.

Proof. Cauchy–Schwarz with (3.16) and (3.17) proves the bounds. For holomorphy, compact interior tests satisfy

$$\int_D M\varphi(V_\lambda f)_y = -\int_D V_\lambda f\,\partial_y(M\varphi).$$

The right side is holomorphic by the stopped estimates. Compact smooth tests are dense in $L^2(M)$, and (3.17) bounds the gradient locally uniformly. The pairing therefore extends holomorphically to every $L^2(M)$ test. Take the tests 1 and M_y/M, the latter admissible by (3.16). The uniform functional bound then gives a norm-convergent Cauchy series. This argument requires no positive lower bound for M at a boundary. □

4. A simple characteristic crossing

We next turn a small-noise cycle calculation into a zero of the full firing-response multiplier. The operator norm comparison and the one-dimensional spectral projection are essential: a small residual alone would not locate a characteristic root of this nonselfadjoint problem.

4.1. **Phase coordinates and the return norm limit.** Recall $X = \log[2/(2-v)]$, $L = \log 2$, and $b(X) = e^X - 1$. Define

$$s(x) = \int_0^x \frac{\mathrm{d}r}{m - b(r)}, \quad \tau = s(L) = \frac{1}{m+1} \log \frac{2m}{m-1}, \quad \omega = \frac{2\pi}{\tau}, \quad Z(s) = \frac{1}{m - b(x(s))}. \tag{4.1}$$

Here τ is the deterministic cycle duration; it differs from the random first hit τ_{f} and from the nonlinear period T used later. Writing $A = m + 1$, direct integration gives

$$Z(s) = \frac{1}{A} + \frac{1}{mA} e^{As}, \qquad \Delta Z = Z(\tau) - Z(0) > 0. \tag{4.2}$$

In (s, η) variables the backward expression is exactly

$$\mathcal{G} = (1 + \varepsilon \eta Z(s))\partial_s + \mathcal{A}. \tag{4.3}$$

Phase endpoints are identified by reset. Functions with equal endpoint values have no reset jump; Z itself has different one-sided values there.

Let Q_t be the OU operator with mean $e^{-t}\eta$ and variance $1 - e^{-2t}$.

Proposition 4.1 (Return comparison in operator norm)**.** *For each compact set Λ in a sufficiently small strip $|\operatorname{Re}\lambda| < r_0$,*

$$\sup_{m \in I,\, \lambda \in \Lambda} \left\| P_a(\lambda) - e^{-\lambda \tau(m)} Q_{\tau(m)} \right\|_{\mathcal{B}_\beta \to \mathcal{B}_\beta} \longrightarrow 0 \qquad (a \downarrow 0). \tag{4.4}$$

The physical return operator, extended constantly below the reflecting boundary as in (3.11)*, maps $\mathcal{B}_\beta$ into itself and is locally norm holomorphic in λ. No rate in* (4.4) *is asserted after the conductance cutoffs have been removed.*

Proof. We give the three estimates and the order of their limits. For $\kappa > 0$,

$$\frac{\mathcal{G}(e^{\kappa X} w_\beta)}{e^{\kappa X} w_\beta} = \kappa(m + \varepsilon\eta - b(X)) + 2\beta - 2\beta(1 - 2\beta)\eta^2.$$

After adding a fixed positive discount exponent this is nonpositive outside a core $|\eta| \le R_\kappa$, uniformly in small ε. Reflection again contributes nonpositively. Stop at the core entrance S or τ_{f}. On $\tau_{\mathrm{f}} < S$, $X_{\tau_{\mathrm{f}}} = L$, so the terminal expectation gains the factor $e^{-\kappa L}$. On $S < \tau_{\mathrm{f}}$, the continued killed expectation is bounded by $C w_\beta(R_\kappa)$, and $\mathbb{E} e^{rS} \mathbf{1}_{\{S < \tau_{\mathrm{f}}\}} \le \mathbb{E} e^{r\tau_{\mathrm{f}}} \le C$. Consequently

$$\sup_{|\eta| \ge M} \frac{|P_a(\lambda) f(\eta)|}{w_\beta(\eta)} \le \left[e^{-\kappa L} + C w_\beta(R_\kappa) e^{-\beta M^2} \right] \|f\|_\beta \tag{4.5}$$

on physical rows. For the constantly extended rows, the estimate at the floor is multiplied by $w_\beta(-m/\varepsilon)/w_\beta(\eta) \le 1$. First choose κ, then M. This makes the weighted row tails small uniformly as $a \downarrow 0$. The comparison operator has the same property, since, with $v_t = 1 - e^{-2t}$,

$$Q_t w_\beta(\eta) = (1 - 2\beta v_t)^{-1/2} \exp\left(\frac{\beta e^{-2t} \eta^2}{1 - 2\beta v_t} \right).$$

For $t = \tau(m)$ its exponent is strictly below β, uniformly on I.

Next fix $|\eta_0| \le M$ and $R > M$. For small enough ε, the floor lies below $-R$ and the voltage speed before leaving $|\eta| < R$ is bounded below by $j_* > 0$. A hit or an exit therefore occurs before $D = 1/j_*$. The explicit unreflected OU representation and the maximal inequality give

$$\mathbb{P}(\text{exit } (-R, R) \text{ before } \tau_{\mathrm{f}}) \le \frac{C_D}{(R-M)^2}.$$

By the stopped estimate with 2β and twice the discount exponent, Cauchy–Schwarz bounds its terminal weighted contribution by a constant depending on M times the square root of this probability. The same argument removes paths exiting $(-R, R)$ for Q_τ. These bounds hold for every datum in the unit ball of $\mathcal{B}_\beta$.

On the fixed interval $(-R, R)$, phase is a positive clock. In reversed phase time the killed evolution is

$$q_\varepsilon(t,\eta)u_t = (\mathcal{A} - \lambda)u, \qquad q_\varepsilon(t,\eta) = 1 + \varepsilon\eta Z(\tau - t), \qquad u|_{\eta=\pm R} = 0, \quad u(0) = f. \tag{4.6}$$

Appendix B constructs this evolution for arbitrary continuous data and proves

$$\|u_\varepsilon(\tau) - u_0(\tau)\|_\infty \le C_{R,I,\Lambda}\varepsilon \|f\|_\infty . \tag{4.7}$$

Thus choose κ, M, then R, and finally ε. On the fixed interval a unit $\mathcal{B}_\beta$ datum has supremum at most $e^{\beta R^2}$; (4.7) now proves (4.4).

For fixed $a > 0$, continuity of the return rows follows from continuous Skorokhod and voltage dependence up to a transverse hit, point-start non-grazing, and higher-weight uniform integrability. This includes the floor, as proved in Appendix A. The constant extension is continuous at that boundary. Estimate (4.5) gives vanishing weighted tails, so the output belongs to $\mathcal{B}_\beta$. The stopped discount series gives norm holomorphy. □

4.2. **The corrected phase mode.** A stationary phase average is useful before constructing the mode. For every continuous τ-periodic function h, let $\overline{h} = \tau^{-1}\int_0^\tau h(s)\,\mathrm{d}s$ and take a periodic primitive $q' = h - \overline{h}$. Along a stationary process q has no reset jump and no reflecting derivative. Therefore

$$\mathbb{E}_M h - \overline{h} = -\varepsilon\mathbb{E}_M[\eta Z(s)(h - \overline{h})], \qquad |\mathbb{E}_M h - \overline{h}| \le C\varepsilon \|h\|_\infty . \tag{4.8}$$

The Gaussian marginal justifies the expectation. This identity avoids any assertion of a joint phase–conductance product limit.

Fix a nonzero integer k and set $q = k\omega$. Choose correctors with equal endpoint values by

$$\begin{aligned} \phi' - \phi &= -iqZ, & \mathcal{W} &= Z(iq\phi + \phi'), & \lambda_2 &= \overline{\mathcal{W}}, \\ \psi' - 2\psi &= -\mathcal{W}, & \chi' &= \lambda_2 - \mathcal{W}, & \overline{\chi} &= 0. \end{aligned} \tag{4.9}$$

The first two periodic first-order equations are uniquely solvable, and the last has a solution because its right side has mean zero. All interval derivatives needed below are bounded for this fixed k, uniformly in $m \in I$. Define initially

$$u_a = e^{iqs}\big[1 + \varepsilon\eta\phi + a((\eta^2 - 1)\psi + \chi)\big]. \tag{4.10}$$

Since $\mathcal{A}\eta = -\eta$ and $\mathcal{A}(\eta^2 - 1) = -2(\eta^2 - 1)$, the coefficients of orders $1, \varepsilon, a$ in $(\mathcal{G} - iq - a\lambda_2)u_a$ vanish exactly. Using a smooth cutoff, make the corrections flat in a fixed-width layer at the physical floor $\eta = -m/\varepsilon$. They remain unchanged above that layer. The resulting u_a has exact Neumann derivative at the floor and zero reset jump. Cutoff errors are polynomial in ε^{-1} times a Gaussian-small weighted tail, hence are exponentially small.

Let the feedback of this smooth field be

$$\alpha = c\int_D M(u_a)_y - c^2\int_D M_y(u_a)_y.$$

Put

$$u_{\mathrm{app}} = u_a + \alpha(s/\tau)e^{iqs}, \qquad \lambda_{\mathrm{app}} = iq + a\lambda_2 + \alpha/\tau, \qquad f_{\mathrm{app}} = (u_{\mathrm{app}})_F. \tag{4.11}$$

The added term has zero y derivative. The reset jump and the feedback of u_{app} both equal α.

Lemma 4.2 (Damping, gain, and residual)**.** *For a fixed $k \neq 0$ and a fixed C_0, uniformly for complex $|c| \le C_0 a$,*

$$\|f_{\mathrm{app}} - \mathbf{1}\|_\beta \le C\sqrt{a}, \qquad \sup_{\text{physical } s,\eta} \frac{|(\mathcal{G} - \lambda_{\mathrm{app}})u_{\mathrm{app}}|}{w_\beta(\eta)} \le Ca^{3/2}. \tag{4.12}$$

There are real $D = D(m) > 0$ and complex $G_k = G_k(m)$ such that

$$\operatorname{Re}\lambda_2 = -k^2 D, \qquad \frac{\alpha}{\tau} = cG_k + O(|c|\sqrt{a} + |c|^2), \tag{4.13}$$

$$G_k = \frac{\Delta Z}{\tau^2}\frac{iq}{(A+iq)(1+iq)}, \qquad \gamma_k := \operatorname{Re} G_k = \frac{\Delta Z}{\tau^2}\frac{q^2(A+1)}{(A^2+q^2)(1+q^2)} > 0. \tag{4.14}$$

Moreover γ_k/k^2 strictly decreases for positive integers k.

Proof. Let the real periodic $\mathcal{C}$ solve $\mathcal{C}' - \mathcal{C} = -Z$. Then $\phi = iq\mathcal{C}$, and

$$\operatorname{Re}\lambda_2 = -q^2\overline{Z\mathcal{C}} = -q^2\overline{\mathcal{C}^2} = -k^2 D, \qquad D = \omega^2\overline{\mathcal{C}^2} > 0.$$

Here $Z = \mathcal{C} - \mathcal{C}'$ and periodicity give the second equality. Positivity follows also from $\mathcal{C}(s) = \int_0^\infty e^{-t} Z(s + t \bmod \tau)\,\mathrm{d}t > 0$.

The leading conductance derivative of (4.10) is $e^{iqs}\phi(s)$, continuous at the phase seam. Formula (4.8) replaces its M average by its phase average with $O(\sqrt{a})$ error. The next derivative term is $\sqrt{a}$ times η times a bounded phase function; the floor error is Gaussian small. With the Fourier convention $Z_j = \tau^{-1}\int_0^\tau Z(s)e^{-ij\omega s}\,\mathrm{d}s$,

$$Z_{-k} = \frac{\Delta Z}{\tau(A+iq)}, \qquad \phi_{-k} = \frac{iqZ_{-k}}{1+iq}.$$

Integration by parts in the diffusion term is valid for the smooth Neumann field and bounds $\int M(u_a)_{yy}$ uniformly. These facts give (4.13). Taking the real part yields (4.14); its displayed denominators prove the strict decrease.

In particular $\alpha = O(a)$. The exact cancellations in (4.9) leave $O(a^{3/2})$ times a fixed polynomial in η. The added term in (4.11) contributes $O(\alpha\varepsilon) + O(\alpha a)$ after its leading frequency shift has canceled. All polynomial terms are bounded in the weighted supremum norm. The floor cutoff contributes exponentially small errors. Extending f_{app} constantly below the reflecting boundary preserves the reset identity. These observations prove (4.12). □

4.3. **An actual root and the full response.** Define the coupled return operator

$$K(\lambda, c) = P_a(\lambda) + \mathbf{1} \otimes \mathfrak{b}_{\lambda,c} \quad \text{on } \mathcal{B}_\beta. \tag{4.15}$$

By Corollary 3.6 and Proposition 4.1, for $|c| \le C_0 a$ it converges in norm to $e^{-\lambda\tau}Q_\tau$ on fixed complex neighborhoods of $ik\omega$.

Proposition 4.3 (The simple return root)**.** *For each fixed $k \neq 0$ and all sufficiently small a, there is a unique root λ_k near $ik\omega$ of the simple eigenvalue-one equation for K. It is simple and holomorphic in c for $|c| < C_0 a$, and*

$$\lambda_k = ik\omega + a\lambda_2 + cG_k + O(a^{3/2}), \tag{4.16}$$

$$\partial_c \lambda_k = G_k + O(\sqrt{a}) \tag{4.17}$$

on interior c disks of radius comparable to a. Constants are uniform in $m \in I$ for fixed k. There are real positive $c_{\mathrm{H}}(a,m)$ and $\Omega_{\mathrm{H}}(a,m)$ satisfying

$$\lambda_1(c_{\mathrm{H}}) = i\Omega_{\mathrm{H}}, \qquad \frac{c_{\mathrm{H}}}{a} = \frac{D}{\gamma_1} + O(\sqrt{a}), \qquad \Omega_{\mathrm{H}} \to \omega, \qquad \operatorname{Re} \partial_c \lambda_1(c_{\mathrm{H}}) > 0. \tag{4.18}$$

Proof. Let $\Pi_\gamma f = (\int f \, \mathrm{d}\gamma)\mathbf{1}$, where γ is standard Gaussian measure. The Mehler formula gives

$$\|Q_{n\tau} - \Pi_\gamma\|_{\mathcal{B}_\beta \to \mathcal{B}_\beta} \le Ce^{-n\tau}. \tag{4.19}$$

For completeness, differentiate its Gaussian kernel in the mean $e^{-t}\eta$ and variance $1 - e^{-2t}$. After multiplication by $e^{-\beta\eta^2}$, the kernel difference is bounded, for large t, by Ce^{-t} times a polynomial in η times $e^{-\beta\eta^2/2}$. Integrating against w_β proves the bound; the finitely many earlier times are absorbed in C. The estimate is uniform for $\tau(m)$ on its compact interval.

It follows that the constant eigenvalue of Q_τ is simple and that the complement resolvent outside $|\zeta| = e^{-\tau}$ is given by a uniformly convergent series. Choose a small circle around 1 in the ζ plane and a sufficiently small fixed λ disk around $ik\omega$. The limiting resolvent on that circle is uniformly bounded. The actual resolvent exists there by a Neumann series. Its contour projection $\Pi = \Pi_{a,c,\lambda}$ tends in norm to Π_γ and has rank one: projections at distance less than one are injective on each other's ranges. Resolvent identities give idempotence and commutation. Normalizing by Gaussian mean gives a holomorphic eigenvalue $e(a,c,\lambda)$ with

$$e \longrightarrow e^{-\lambda\tau}, \qquad e_\lambda \longrightarrow -\tau$$

at the harmonic; derivative convergence on a smaller disk follows from Cauchy's formula.

At λ_{app} let $v = V_{\lambda_{\mathrm{app}}} f_{\mathrm{app}}$ and $w = u_{\mathrm{app}} - v$. It has zero outgoing terminal value and source $r = -(\mathcal{G} - \lambda_{\mathrm{app}})u_{\mathrm{app}}$ bounded by $Ca^{3/2}w_\beta$. This source is smooth on the closed physical half-strip: the phase correctors are smooth up to their separate voltage endpoints, and the floor cutoff is smooth. Thus it belongs to the source class of Lemma 3.5. The killed-reward bound and the forced version of (3.19) therefore give

$$(I - K(\lambda_{\mathrm{app}}, c))f_{\mathrm{app}} = w(0,\cdot) + \mathfrak{b}_{\mathrm{field}}(w)\mathbf{1}, \qquad \|(I-K)f_{\mathrm{app}}\|_\beta \le Ca^{3/2}.$$

Here $\mathfrak{b}_{\mathrm{field}}(w) = c\int Mw_y - c^2 \int M_y w_y$. Since $\Pi f_{\mathrm{app}} \to \mathbf{1}$, projecting this identity yields $|e(\lambda_{\mathrm{app}}, c) - 1| \le Ca^{3/2}$.

On a smaller fixed convex disk choose $|e_\lambda + \tau| < \tau/3$; $e_{\lambda\lambda}$ is bounded there. With $b = e_\lambda(\lambda_{\mathrm{app}}, c)$, the map

$$\delta \longmapsto \delta - \frac{e(\lambda_{\mathrm{app}} + \delta, c) - 1}{b}$$

is a contraction on $|\delta| \le C' a^{3/2}$ for sufficiently large fixed C'. Its value at zero is $O(a^{3/2})$, and its Lipschitz constant on that disk is $O(a^{3/2})$. It gives an actual root with (4.16). If two roots lay in the fixed convex disk, integrating $e_\lambda + \tau$ on the segment between them would give $\tau|\lambda - \lambda'| \le (\tau/3)|\lambda - \lambda'|$. Thus the root is unique there and simple.

Holomorphy and Cauchy's formula on c disks of radius comparable to a turn the uniform $O(a^{3/2})$ error into the $O(\sqrt{a})$ derivative error in (4.17). Choose $C_0 > 4 \sup_I D/\gamma_1$. On a real interior interval of c of length $C_1 a$, the real part of λ_1 starts negative, ends positive, and has strictly positive derivative. Its unique zero gives (4.18). This concerns the indicated branch; no stability assertion about the remaining spectrum is required. □

We now compare this root with the physical forward response. Define

$$g_c = -cM + c^2 M_y, \qquad \mathcal{H}(\lambda; m, a, c) = \ell R_\lambda B g_c. \tag{4.20}$$

The response is defined on the entire imaginary axis and locally holomorphic near each frequency by Proposition 2.3.

Proposition 4.4 (Response crossing and the static block)**.** *At the crossing in* (4.18)*,*

$$\mathcal{H}(i\Omega_{\mathrm{H}}; m, a, c_{\mathrm{H}}) = 1, \qquad \partial_\lambda(1-\mathcal{H})(i\Omega_{\mathrm{H}}; m, a, c_{\mathrm{H}}) \neq 0. \tag{4.21}$$

For $c = O(a)$,

$$\mathcal{H}(0; m, a, c) = O(\sqrt{a}). \tag{4.22}$$

Proof. Near the chosen nonzero harmonic $I - P_a(\lambda)$ is Fredholm of index zero. Indeed the norm limit splits into a one-dimensional part and a complement on which $I - P$ is invertible; block elimination leaves a finite-dimensional block. On the nonzero imaginary axis its kernel is zero. If $(I - P)f = 0$, then $u = V_\lambda f$ has zero reset jump. Proposition 3.4 puts u in the actual transpose domain. Its M mean is zero since $\lambda \neq 0$, and the frozen gap excludes this imaginary eigenfunction. Thus $I - P$ is invertible at the crossing and in a possibly a-dependent complex neighborhood.

Set $f_0 = (I - P)^{-1}\mathbf{1}$. Its killed value has jump one. The completed Green pairing, with both feedback terms, gives

$$\mathcal{H}(\lambda) = \mathfrak{b}_{\lambda,c}(f_0), \qquad (I - K)f_0 = (1 - \mathcal{H}(\lambda))\mathbf{1}. \tag{4.23}$$

At a K root on this imaginary axis the jump of an eigenfunction cannot be zero, by the same frozen transpose argument. The first identity follows from (3.14); the second therefore implies $\mathcal{H} = 1$.

Let ν be the left eigenfunctional of the simple eigenvalue of K, normalized by $\nu(f_0) = 1$ at the root. It satisfies $\nu(I - P) = \nu(\mathbf{1})\mathfrak{b}$. If $\nu(\mathbf{1}) = 0$, invertibility of $I - P$ would force $\nu = 0$. Differentiating (4.23) and pairing with ν thus gives

$$-e_\lambda = \partial_\lambda(1 - \mathcal{H})\,\nu(\mathbf{1}).$$

This proves (4.21), including simplicity.

At zero frequency, $P_a(0)\mathbf{1} = \mathbf{1}$. Its simple spectral projection is $\Pi_a f = \pi_a(f)\mathbf{1}$, by the invariant mark law. The inverse of $I - P_a(0)$ on π_a-mean-zero data is uniformly bounded, by the same separated complement. Let $t(z) = \mathbb{E}_z\tau_{\mathrm{f}}$, with reset value t_0. The occupation formula gives $\pi_a(1 - N_0 t_0) = 0$. Solve

$$(I - P_a(0))f = 1 - N_0 t_0, \qquad \pi_a f = 0, \qquad w = V_0 f - N_0 t.$$

Here t has zero terminal datum and source $r = 1$, while w has source $r = -N_0$ at $\lambda = 0$. These constant sources satisfy the smoothness and weighted bounds in Lemma 3.5. Then $\mathcal{G}w = N_0$ and $w_F - w_0 = 1$. Killed-reward and forced cycle energy estimates give $\int M|w|^2 + a\int M|w_y|^2 \leq C$. The polynomial-growth argument in Appendix A also applies with this bounded interior source. The generator pairing now has the constant volume term $N_0 \int h$. It vanishes for $h = R_0 B g_c$, whose mass is zero. The same Riesz completion therefore gives $\mathcal{H}(0) = c\int M w_y - c^2 \int M_y w_y = O(\sqrt{a})$. For small a, the static multiplier $1 - \mathcal{H}(0)$ is nonzero. □

5. Periodic rates and nonnegative full densities

The selected first characteristic mode does not by itself produce a periodic density. We first remove all other harmonic resonances, solve a scalar periodic rate equation, and finally identify its signed linear response with a positive prescribed evolution.

5.1. **Physical parameter dependence and harmonic selection.**

All parameter perturbations in this section use the original physical variables on the unchanged half-strip. In particular the floor does not move when m or a changes.

Lemma 5.1 (Analyticity at positive noise)**.** *Locally at every $a > 0$ under consideration, the completed inverse $R_{i\theta}B : \mathsf{H}_\sigma \to \mathsf{E}_\sigma$ is holomorphic in (m, a), uniformly in $\theta \in \mathbb{R}$ on a sufficiently small complex parameter neighborhood. The stationary triple (M, M_y, M_F) and $N_0 = \ell M$ are locally analytic. The response* (4.20) *is locally analytic in (λ, c, m, a) near every imaginary frequency. The selected crossing functions $c_{\mathrm{H}}, \Omega_{\mathrm{H}}$ are real analytic on $U = (0, a_0) \times (2, 3)$.*

Proof. At a fixed base (m, a), regard the density, derivative, and outgoing trace as separate components of E_σ. A perturbed inverse solves

$$h = R_{i\theta} B[g - \delta_m h + \delta_a h_y]. \tag{5.1}$$

The right side uses the first and second components of h. The uniform triple bound makes its linear perturbation small on the ambient product for small complex δ_m, δ_a, independently of θ. Its Neumann series is analytic. Every output has its derivative and trace identified by the completed weak equation; closedness of the distributional derivative and passage in that equation retain these identities in the limit. Its floor flux is the perturbed natural flux. For real parameters, homogeneous uniqueness identifies it with the actual prescribed inverse.

Likewise the nearby normalized stationary density is $M + h$, where

$$h = R_0 B[-\delta_m(M + h) + \delta_a(M_y + h_y)], \qquad \int_D h = 0. \tag{5.2}$$

The same series gives analyticity. For real parameters normalized stationary uniqueness identifies this density with the stationary state supplied by Proposition 2.1, including its positivity. Weighted norm closeness alone is not used for that conclusion. Local frequency analyticity follows from the ordinary resolvent identity as in Proposition 2.3. This proves the response statement.

The simple zero and the nonzero real coupling derivative in (4.18) and (4.21) give analytic crossing functions by the implicit function argument. Local solutions patch on U by uniqueness of the selected branch and its crossing. No uniform radius in m as $a \downarrow 0$, and no convergence of parameter derivatives in that limit, is asserted. □

Lemma 5.2 (A base point with only first-harmonic resonance)**.** *There is $(a, m_*) \in U$ such that, writing*

$$c_* = c_{\mathrm{H}}(a, m_*), \qquad \Omega_* = \Omega_{\mathrm{H}}(a, m_*), \qquad T = 2\pi/\Omega_*,$$

one has

$$\partial_m \Omega_{\mathrm{H}}(a, m_*) \neq 0, \qquad 1 - \mathcal{H}(ik\Omega_*; m_*, a, c_*) \neq 0 \quad (k \in \mathbb{Z} \setminus \{-1, 1\}). \tag{5.3}$$

The first-harmonic zeros are simple.

Proof. For each fixed $k \geq 2$, Proposition 4.3 gives, at $c = c_{\mathrm{H}}$,

$$\operatorname{Re} \lambda_k = a \left[-k^2 D + \frac{D}{\gamma_1} \gamma_k \right] + O(a^{3/2}) < 0$$

for small a depending on this k. The strict sign follows from $\gamma_k/k^2 < \gamma_1$. The real analytic complex-valued function

$$r_k(a, m) = 1 - \mathcal{H}(ik\Omega_{\mathrm{H}}(a, m); m, a, c_{\mathrm{H}}(a, m))$$

is not identically zero on U. Indeed, for this fixed k take a small enough that $ik\Omega_{\mathrm{H}}$ lies in its fixed return-root disk. A zero of r_k would, by the imaginary-axis matching in (4.23), give an eigenvalue-one equation for K there. Its only root is λ_k, which has negative real part, a contradiction.

Also $\Omega_{\mathrm{H}}(a, m) \to \omega(m)$ uniformly on I, while

$$\tau_m = -\int_0^L [m - b(x)]^{-2} \, \mathrm{d}x < 0.$$

Choose two interior means with different ω. For all sufficiently small a, $\Omega_{\mathrm{H}}(a, \cdot)$ has different values at those means. Thus $\partial_m \Omega_{\mathrm{H}}$ is not the zero analytic function on U. The zero set of a nonzero real analytic function on this connected open set has empty interior: vanishing on a ball makes all its power-series coefficients zero and propagates along a chain of overlapping balls. For complex-valued r_k, apply this observation to a nonzero real or imaginary part.

The zero sets of all r_k and of $\partial_m \Omega_{\mathrm{H}}$ are closed with empty interior in U. Successively choose nested closed balls, each inside the preceding interior, avoiding the next zero set, with radii tending

to zero and the first ball compactly inside U. Completeness gives a point avoiding every set. The static multiplier is nonzero by (4.22); negative harmonics are complex conjugates. This proves (5.3). Simplicity at the first harmonics follows from (4.21). □

The selection is existential and uses only a fixed-k asymptotic for each integer k. It does not require a small-noise expansion uniform over infinitely many harmonics.

5.2. **The periodic inverse and the rate equation.** From now on fix the point and the period in Lemma 5.2. Use Fourier coefficients $u_k = T^{-1}\int_0^T u(t)e^{-ik\Omega_* t}\,\mathrm{d}t$, so that

$$\|u\|^2_{H^1(0,T;E)} = T\sum_{k\in\mathbb{Z}}(1+k^2\Omega_*^2)\,\|u_k\|_E^2$$

for a Hilbert space E.

Lemma 5.3 (High-frequency response and periodic inverse)**.** *At fixed positive parameters,*

$$\mathcal{H}(i\theta; m, a, c) \longrightarrow 0 \quad (|\theta|\to\infty). \tag{5.4}$$

The coefficientwise map

$$(\mathcal{R}_T B g)_k = \mathcal{R}_{k\Omega_*} g_k$$

is bounded from $H^1_{\mathrm{per}}(0,T;\mathsf{H}_\sigma)$ to $H^1_{\mathrm{per}}(0,T;\mathsf{E}_\sigma)$, with its density, gradient, trace, and floor identities. The linear rate-response multiplier on $H^1_{\mathrm{per}}(0,T)$ is compact.

Proof. For fixed $h\in\mathsf{H}_{\sigma,0}$, $t\mapsto S(t)h$ is Bochner integrable by the frozen gap. Approximate it in $L^1_t\mathsf{H}_\sigma$ by finitely many step functions and integrate their scalar exponentials. Formula (2.9) then shows $R_{i\theta}h\to 0$ in H_σ. Apply (2.7) to this resolvent:

$$\|(R_{i\theta}h)_F\|^2_{\mathsf{O}_\sigma} \le C_a\,\|R_{i\theta}h\|^2_{\mathsf{H}_\sigma} + C_a\,\|h\|_{\mathsf{H}_\sigma}\,\|R_{i\theta}h\|_{\mathsf{H}_\sigma}\,.$$

Its outgoing trace tends to zero as well. Approximate g_c in H_σ by compact smooth g_j supported away from the floor. For fixed j, Bg_j is an ordinary zero-mass datum, so the preceding argument applies. The uniform completed trace estimate controls the approximation error independently of θ, proving (5.4).

Parseval's identity and (2.12) give the stated periodic bound first on finite Fourier sums, then by completion. The completed weak equation identifies the limiting derivative and traces, preserves mass zero, and retains the natural floor input. The density belongs to $C_{\mathrm{per}}(0,T;\mathsf{H}_\sigma)$. Finally the scalar multipliers tend to zero by (5.4), so finite Fourier truncations approximate the linear response in operator norm on the same H^1 space. □

Keep a fixed and allow c, m near c_*, m_*. For a small real $n\in H^1_{\mathrm{per}}(0,T)$ solve the completed equation

$$h_n = R_T B\{n g_c + n[-c h_n + c^2 (h_n)_y]\}, \qquad \int_D h_n(t) = 0. \tag{5.5}$$

Here and below $R_T B$ denotes the density component when the entire triple is understood. The one-dimensional Sobolev product inequality and Lemma 5.3 bound the last term on $H^1_t\mathsf{E}_\sigma$ by $C\,\|n\|_{H^1}$ times the triple norm. A Neumann series therefore gives a unique small analytic solution. Parameter analyticity follows uniformly in Fourier frequency from (5.1). Put $F_n = M + h_n$. Its weak equation has precisely the prescribed mean $m + cn$ and diffusion $a + c^2 n$, with the voltage traces and the natural floor flux. At this stage F_n is allowed to be signed.

The actual self-consistency equation is the scalar equation

$$\mathcal{F}(n, c, m) := n - \ell h_n = 0. \tag{5.6}$$

Its linearization at $n = 0$ is the Fourier multiplier

$$d_k = 1 - \mathcal{H}(ik\Omega_*; m, a, c).$$

At the base point only $k = \pm 1$ are singular. Its restriction to the complement is boundedly invertible: finitely many nonzero d_k are bounded away from zero, and (5.4) gives $|d_k| \geq 1/2$ in the remaining tail.

Proposition 5.4 (A nonconstant self-consistent rate)**.** *For all sufficiently small $r > 0$ there are real analytic parameter values $c(r^2), m(r^2)$ near c_*, m_* and real $n_r \in H^1_{\mathrm{per}}(0,T)$ solving (5.6), with exact first Fourier coefficient $(n_r)_1 = r$ and $\|n_r\|_{H^1} \to 0$.*

Proof. Write $n = ze^{i\Omega_* t} + \overline{z}e^{-i\Omega_* t} + w$, with $w_{\pm 1} = 0$. The complementary equation has an analytic solution $w = w(z, \overline{z}, c, m)$ by its bounded inverse and the local implicit function argument. Equivalently, subtract its linear part and use the resulting contraction. Time translations commute with the equation and its projections, so uniqueness makes w translation covariant. In the remaining first coefficient, a real analytic monomial $z^p \overline{z}^q$ must have $p - q = 1$. Consequently

$$\mathcal{F}_1 = z\, g(|z|^2, c, m), \qquad g(0, c, m) = 1 - \mathcal{H}(i\Omega_*; m, a, c). \tag{5.7}$$

The latter equality holds also for nearby parameters: the linear rate operator remains diagonal in Fourier modes, so its first-mode input has zero complementary projection. The persistent complementary inverse thus gives $w_z(0) = 0$.

Let $\lambda(c, m)$ be the simple characteristic root near $i\Omega_*$. Along its crossing curve,

$$\lambda_m + \lambda_c\, \partial_m c_{\mathrm{H}} = i\, \partial_m \Omega_{\mathrm{H}}, \qquad \det_{\mathbb{R}}(\lambda_c, \lambda_m) = \operatorname{Re} \lambda_c\, \partial_m \Omega_{\mathrm{H}} \neq 0.$$

The determinant uses the real and imaginary parts as its two rows. If $d(\lambda, c, m) = 1 - \mathcal{H}(\lambda; m, a, c)$, differentiation of $d(\lambda(c, m), c, m) = 0$ multiplies these two parameter columns by the same nonzero complex number $-d_\lambda$. It therefore preserves nonsingularity. The real Jacobian of $g(0, c, m)$ in (c, m) is invertible. The finite-dimensional analytic implicit function argument now gives $c(\rho), m(\rho)$ with $g(\rho, c(\rho), m(\rho)) = 0$ for small real ρ. Take $\rho = r^2$, $z = r > 0$. This proves the proposition. The use of two free physical parameters avoids any assumption about a cubic coefficient in a one-parameter bifurcation. □

5.3. **Positivity and the physical solution.** An $H^1_t \mathsf{H}_\sigma$ perturbation of M need not be pointwise nonnegative. The next argument uses a positive period map instead.

Lemma 5.5 (Comparison of prescribed evolutions)**.** *On each fixed finite interval $[0, t_*]$, let $S_n(t, 0)$ be the prescribed positive conservative evolution with mean $m + cn(t)$ and diffusion $a + c^2 n(t)$. For small real n and nearby m, c,*

$$\|S_n(t_*, 0) - S_*(t_*)\|_{\mathsf{H}_\sigma \to \mathsf{H}_\sigma} \leq C_{a,t_*}\big(|m - m_*| + \|n\|_\infty\big), \tag{5.8}$$

where S_* is the frozen evolution at (m_*, a).

Proof. For the same initial datum f put $u_n = S_n(\cdot, 0)f$. The homogeneous forward energy is uniform on a small positive coefficient neighborhood, so $\|u_n\|_{L^\infty_t \mathsf{H}_\sigma} + \|(u_n)_y\|_{L^2_t \mathsf{H}_\sigma} \leq C_{a,t_*} \|f\|_{\mathsf{H}_\sigma}$. The difference $u_n - S_* f$ solves the base forced equation with zero initial datum and input

$$g = -[(m - m_*) + cn]u_n + c^2 n (u_n)_y.$$

Its $L^2_t \mathsf{H}_\sigma$ norm is at most the right side of (5.8) times $\|f\|$. The completed weak flux identity and prescribed uniqueness identify this difference with the solution in Lemma 2.4. Estimate (2.13) proves the assertion. This uses a causal forcing estimate, not an assumed forced renormalization formula. □

Proof of Theorem 1.1. Choose the base point in Lemma 5.2. By the frozen gap, there is an integer p with $\|S_*(pT)\|_{\mathsf{H}_{\sigma,0} \to \mathsf{H}_{\sigma,0}} < 1/4$. Keep this integer fixed. For the solution in Proposition 5.4, choose $r > 0$ so small that (5.8) at pT is less than $1/4$. The prescribed pT map then contracts every mass-zero difference by a factor less than $1/2$.

Iterate that map from the normalized nonnegative density M. Every iterate is nonnegative and has mass one. The affine mass-one space in H_σ is complete, so the iterates converge to its unique fixed point F_0^*. Positivity and mass one pass to the limit, since their defining sets are closed. If $P_n = S_n(T,0)$, periodic coefficients and uniqueness give $S_n(pT,0) = P_n^p$. The operator P_n commutes with this power, and $P_n F_0^*$ is another mass-one fixed point of it. Uniqueness gives $P_n F_0^* = F_0^*$. Thus $F^*(t) = S_n(t,0)F_0^*$ is nonnegative and T-periodic.

The signed response $F_n = M + h_n$ in (5.5) lies in $C_t\mathsf{H}_\sigma$ with its derivative and traces in the prescribed energy class. Its completed weak equation is the same prescribed homogeneous equation. Prescribed uniqueness identifies it with its evolution from $F_n(0)$. Since it is T-periodic and of mass one, $F_n(0)$ is a fixed point of the pT map on the full affine space, and hence equals F_0^*. This proves $F_n = F^* \geq 0$ without a lower bound for M.

Set, with $m = m(r^2)$ and $c = c(r^2)$,

$$y_* = m - cN_0(m,a), \qquad a_* = a - c^2 N_0(m,a). \tag{5.9}$$

At the base point $c = O(a)$ and N_0 is uniformly bounded, while $m > 1$. First choose a small and then r small. This makes y_*, a_*, c positive. The exact rate equation gives $N = N_0 + n_r$, so its first Fourier coefficient is $r > 0$. Moreover

$$y_* + cN = m + cn_r, \qquad a_* + c^2 N = a + c^2 n_r.$$

The prescribed equation for F_n is therefore the original nonlinear equation with both feedback terms. The choices $v_E = 2, v_F = 1, y_L = 1$ satisfy all remaining physical parameter inequalities.

We verify the requested solution properties. The density belongs to $H^1_{\mathrm{per}}(0,T;\mathsf{H}_\sigma)$, hence to $C_{\mathrm{per}}(0,T;\mathsf{H}_\sigma) \subset C_{\mathrm{per}}(0,T;L^1(D))$. The continuous representative has mass one at every time. Exponential-weight Cauchy–Schwarz bounds all finite conductance moments uniformly in time, in particular $\int y^2 F$. The positive entropy part is bounded by $\int (F^2 + F)$. For the negative part choose the normalized density $q_0(v,y) = Ce^{-y^2}$. The pointwise inequality $s\log(s/q_0) \geq s - q_0$, together with the second moment and the finite positive part, bounds $\int |F \log F|$ uniformly over the period, with $0 \log 0 = 0$.

The low-conductance zero traces and high-conductance flux reset are the strong completed voltage traces in (5.5), equivalently those of the prescribed evolution. Its floor trace is zero for

$$Q = (m + cn_r - y)F - (a + c^2 n_r)F_y = (y_* + cN - y)F - (a_* + c^2 N)F_y.$$

Finally pair Q with any smooth periodic time test and any smooth test on the closed voltage interval. This pairing lies in $L^1(\,\mathrm{d}y)$ by the weighted energy bounds. The PDE, integration by parts in t, v, and the integrable outgoing and reset traces put its distributional y derivative in L^1 on every tail. It has an absolutely continuous tail representative with a finite limit at infinity. Its own L^1 integrability forces that limit to be zero. This is the vanishing distributional flux trace in the theorem. All required boundary, entropy, continuity, and moment properties follow for the nonnegative full density $F = F_n$. □

Appendix A. Process identification and the reflecting boundary

This appendix supplies the boundary facts used in Sections 2–3. The interior regularity theorem is used only after a distributional equation has been established. The reflecting condition is constructed separately.

A.1. **From the forward density to the reset process.**

Proof of Proposition 3.1. Let $F_t = S(t)h$. Put $\Phi(v) = (\cos(2\pi v), \sin(2\pi v))$ and lift F_t to a probability ν_t on $\mathbb{R}^3$: choose (v,y) with density F_t and an independent fair sign, then set $(\xi, z) = (\Phi(v), \pm y)$.

This probability is supported on the unit circle times $\mathbb{R}$ and need not have an ambient density. Away from the circle cut $p = (1, 0)$ use the inverse angle $v(\xi) \in (0, 1)$ and define

$$B_\xi = 2\pi J(v(\xi), |z|)(-\xi_2, \xi_1), \qquad B_z = m \operatorname{sign} z - z, \qquad A = \operatorname{diag}(0, 0, 2a).$$

Choose any finite Borel angular drift at the cut and extend it by zero off the circle. The value at the cut does not affect the forward equation, since ν_t gives it zero mass.

For a compact smooth ambient test Ψ, its even pullback is

$$\varphi(v, y) = \tfrac{1}{2}[\Psi(\Phi(v), y) + \Psi(\Phi(v), -y)].$$

It has equal voltage endpoint values and $\varphi_y(v, 0) = 0$. The reset terms cancel in the forward Green identity and the low-conductance traces vanish. Integration by parts in y has no remaining floor contribution. The result is precisely the Euclidean Fokker–Planck equation with drift B and covariance A, whose second-order term is $(1/2)A : D^2 = a\partial_{zz}$. The lifted probabilities are narrowly continuous and

$$\int_0^t \int_{\mathbb{R}^3} (|B| + |A|) \, \mathrm{d}\nu_s \, \mathrm{d}s < \infty,$$

since $|B| \le C(1 + |z|)$ on their support and F has a bounded first conductance moment.

Trevisan's superposition theorem [8, Definition 2.2, condition (4), and Theorem 2.5] supplies a continuous martingale solution with these one-time laws. The angular component lies on the circle at all rational times, and hence at all times by continuity. Localized coordinate and product tests show that its angular martingale has zero quadratic variation, while its signed conductance satisfies

$$\mathrm{d}Z_t = (m \operatorname{sign} Z_t - Z_t) \, \mathrm{d}t + \sqrt{2a} \, \mathrm{d}W_t. \tag{A.1}$$

Here W is scalar Brownian motion in the full solution filtration, by the martingale quadratic variation and the scalar Lévy characterization. The stochastic integral, maximal inequality, localized Itô formula, and scalar characterization used here can, for example, be taken from [3, Theorems 2.32, 4.10, 4.17, 4.19–4.20 and the scalar case of Theorem 5.11].

We justify folding directly. Stop at $\tau_R = \inf\{t : |Z_t| \ge R\}$ and apply the localized Itô formula to $\varphi_\delta(z) = \sqrt{z^2 + \delta^2}$, made bounded outside $[-R - 1, R + 1]$. For $0 < \delta < 1$,

$$a\mathbb{E} \int_0^{t \wedge \tau_R} \varphi_\delta''(Z_s) \, \mathrm{d}s \le 2R + 2 + (m + R)t.$$

On $|z| \le \delta$ its second derivative is at least $(2\sqrt{2}\,\delta)^{-1}$. Hence

$$\mathbb{E} \int_0^{t \wedge \tau_R} \mathbf{1}_{\{|Z_s| \le \delta\}} \, \mathrm{d}s \le C_{a,R,t}\delta.$$

Let $\delta \downarrow 0$ and then $R \uparrow \infty$ to conclude that Z spends zero time at zero. The drift terms in the same Itô formula converge to those for $|Z|$; the stochastic integrals converge uniformly in probability by isometry and the maximal inequality. Consequently

$$K_t = |Z_t| - |Z_0| - \int_0^t (m - |Z_s|) \, \mathrm{d}s - \sqrt{2a} \int_0^t \operatorname{sign}(Z_s) \, \mathrm{d}W_s$$

is a continuous nondecreasing limit of $a \int_0^t \varphi_\delta''(Z_s) \, \mathrm{d}s$. On a path interval where $|Z| \ge d > 0$, the latter increments are bounded by $at\delta^2/d^3$ and vanish. Thus K increases only when $|Z| = 0$. The martingale $W_t' = \int_0^t \operatorname{sign}(Z_s) \, \mathrm{d}W_s$ has quadratic variation t, so it is scalar Brownian motion in the same filtration. The folded $Y = |Z|$ satisfies (3.1).

The half-line Skorokhod map $x(t) \mapsto x(t) + \sup_{s \le t}(-x(s))_+$ has supremum-norm Lipschitz constant at most two. Its integral equation with drift $m - Y$ is a contraction on any interval of length less than $1/2$. Iteration proves pathwise uniqueness and makes Y a measurable function of Y_0, W'. Brownian increments are independent of the initial full sigma-field. If $p(y) = \int_0^1 h(v, y) \, \mathrm{d}v$, Fubini therefore

shows that the conditional density of V_0 given the entire future Y path is $h(v, Y_0)/p(Y_0)$ on the set $p(Y_0) > 0$.

For a fixed continuous conductance path, before a reset,

$$V_t = A_t V_0 + C_t, \quad A_t = \exp\left[-\int_0^t (Y_s + 1)\,\mathrm{d}s\right], \quad C_t = \int_0^t \exp\left[-\int_s^t (Y_r + 1)\,\mathrm{d}r\right] 2Y_s\,\mathrm{d}s. \tag{A.2}$$

The C^1 function $H(t) = (1 - C_t)/A_t$ satisfies $H'(t) = -J_F(Y_t)/A_t$. A grazing threshold hit would make V_0 a critical value of H. Critical values of a C^1 real function on a bounded interval have Lebesgue measure zero: cover the zero-derivative set by open intervals where $|H'| < \epsilon$ and bound the lengths of their images by ϵ times their total length, then let ϵ decrease to zero. Exhaust the time intervals. The conditional density just proved excludes such a hit. A first hit from below has $J_F \geq 0$, and thus in fact $Y > 1$.

After a reset the initial voltage is zero, so a separate point-start argument is required. Until a floor encounter the auxiliary unreflected conductance and voltage flow have a joint density at every positive time. Indeed Itô's formula gives their distributional forward equation. The smooth fields ∂_y and $\partial_t + J\partial_v + (m - y)\partial_y$, and their bracket $(2 - v)\partial_v - \partial_y$, span all time–space directions for $v < 2$. Hörmander's theorem [6, Theorem 1.1] gives a smooth space–time density there. The explicit flow stays below 2; narrow continuity against compact tests upgrades the almost-every-time density identification to every positive time.

Starting after a reset with $Y > 1$, wait $\delta = 1/n$ before the next hit or first floor encounter. The law at δ restricted to this survival event is dominated by the auxiliary law and has a joint density. Future Brownian increments are independent of the past event. The conditional-density critical-value argument therefore applies after δ. The countable collection of these events covers all possible finite next hits. Independence after stopping times follows by approximating them from above by dyadic stopping times and using Brownian continuity and deterministic-time independence. Thus the argument repeats after every reset.

Before its first cut encounter the circle path inverts to the ordinary voltage equation. It cannot reach zero from inside, since $V_t \geq V_0 A_t > 0$. At a non-grazing outgoing hit the angular speeds on both sides of the cut are positive, as $J_F > 0$ and $J_0 = 2Y > 0$. Continuity of Y makes them bounded away from zero in a neighborhood. The path crosses in the positive direction and enters the unique voltage trajectory from zero. It cannot stick on an interval; in addition its total occupation of the cut is zero by the one-time marginals. The pathwise identity for X gives (3.15). Since Y is continuous, there is no finite-time accumulation of hits. The same bound and the first conductance moment make the expected count finite. This identifies the circle process with the reset process, proving its law assertion.

For smooth q supported in $(0, t) \times (1, \infty)$ apply piecewise Itô to $\varphi(s, v, y) = vq(s, y)$, whose reset jump is q. Comparing with the forward Green formula gives (3.2) against these tests. The trace of a nonnegative forward density is nonnegative in its kinetic trace measure; this also follows by the bounded negative-part renormalization in the same Green identity. Both measures are therefore locally finite positive measures. All hits have $Y > 1$. A spike at a deterministic time would put the circle path at the cut, which its marginal excludes. There is no initial spike for a density. Exhaust the conductance and time endpoints. Finite expected count controls the total measure, and monotone convergence extends equality first to bounded Borel tests and then to integrable tests. This proves the complete statement. □

A.2. **Local backward equations, including the floor.** We give the local construction for killed values with the weights of Section 3. Their interior sources belong to the smooth class specified before Lemma 3.5 and obey the stated Gaussian weighted bound. We also treat bounded full-process resolvents with compact smooth source. In both cases write

$$\mathcal{G}u = \lambda u - r. \tag{A.3}$$

The source is zero for a killed harmonic. Nonzero killed sources used below are the constant source in the static block and the smooth corrected-mode residual. The full-process case has $\operatorname{Re}\lambda > 0$ and compact smooth bounded r.

First establish the interior distributional equation. On an interior rectangle use the unreflected, unreset process until exit from a larger rectangle. Its conductance is explicit and its voltage is (A.2); the stochastic flow is triangular in the initial variables, with positive Jacobian $e^{-t}A_t$. For a compact smooth initial density φ, Taylor expansion of its inverse flow and inverse Jacobian gives

$$P_h^T\varphi = \varphi + h[-\partial_v(J\varphi) - \partial_y((m-y)\varphi) + a\varphi_{yy}] + o(h) \tag{A.4}$$

in local L^1 and in the fixed Gaussian-weighted L^1 used to pair with u. To see the remainder explicitly, the conductance increment has mean $(m-y)h + O(h^2)$, variance $2ah + O(h^2)$, and jth absolute moment $O(h^{j/2})$; its random contribution to the voltage increment is $O(h^{3/2})$ in every fixed finite moment. The inverse Jacobian is $1 + h(y+2) + O_{L^p}(h^{3/2})$. Bounded third derivatives of φ control the Taylor remainder. Outside an enlarged compact set the transported density has a Gaussian small-time tail, which dominates the fixed exponential-quadratic weight. The probability of leaving the larger rectangle before h is $O(e^{-c/h})$; higher stopped weights make its contribution $o(h)$. For a smooth killed source the reward increment is $hr + o(h)$ on compact sets of starts. The same localization and the higher-weight reward bounds justify this term in the paired limit. Dynamic programming, pairing with φ and division by h, now give (A.3) in distributions.

We use the real-coefficient regularity theorem literally at complex frequency. Introduce a real variable z and set $U(z,v,y) = e^{-\lambda z}u(v,y)$. Then $(\partial_z + \mathcal{G})U = -e^{-\lambda z}r$. The fields ∂_y and $\partial_z + J\partial_v + (m-y)\partial_y$, with the same bracket $(2-v)\partial_v - \partial_y$, span all three directions. Apply [6, Theorem 1.1] to the real and imaginary parts of U separately. Thus u is interior smooth.

At $v = 0, y > 0$ extend the auxiliary voltage a fixed small distance below zero. An interior physical path never exits through zero; the high-conductance block argument still bounds the hitting time from the added finite voltage interval. The weak equation and interior regularity on this extension make $u(0,y)$ smooth for $y > 0$. For a constant reset jump, the outgoing datum is therefore smooth on $y > 1$. Its positive outgoing speed gives the terminal trace. At $v = 1, y < 1$, choose a small rectangle straddling that face on which $J \le -j_0 < 0$. Stop the auxiliary process on its exit. Its exit time is bounded by the voltage width divided by j_0. Retain the original data on reachable physical sides and extend side data continuously to the added part. A physical start cannot exit through the added upper voltage side. The local value agrees with u inside, solves the weak equation across the face, and is smooth there. The deterministic exit bound suffices for any fixed complex discount; no global hitting estimate outside the physical voltage interval is assumed.

Point starts at positive conductance have no grazing hit by the positive-time density argument above. For a start at $y = 0$, before any finite threshold hit the conductance first reaches $1/2$ strictly earlier; start the same argument at that stop. Continuous dependence of the Skorokhod and voltage equations then implies almost-sure continuity of $\tau_{\mathrm{f}}, Y_{\tau_{\mathrm{f}}}$ in the initial state at a transverse hit. Before the hit the voltage on each shorter compact interval stays a positive distance below threshold, and the final transverse crossing persists. The stopped 4β weights and stronger discount moment give uniform integrability on compact sets of starts. Thus killed values are continuous, including at the floor. For a bounded full-process source, the same coupling on finite intervals and the discounted tail prove continuity.

Fix $0 < v_- < v_+ < 1$ and choose $y_c > 0$ so that $q = -J$ satisfies $0 < q_- \le q \le q_+$ on the resulting floor rectangle. The backward equation becomes the forward voltage-clock problem

$$qu_v = au_{yy} + (m-y)u_y - \lambda u + r, \qquad u_y(v,0) = 0. \tag{A.5}$$

Give it the continuous initial data at v_- and the upper-side data at y_c. After a smooth lifting of that side, use a mixed Neumann–Dirichlet basis on $(0, y_c)$. Its q-weighted mass matrix is uniformly

positive. Testing the finite systems by their solutions yields

$$\tfrac{1}{2}\partial_v \int_0^{y_c} q|w|^2 \,\mathrm{d}y + a\int_0^{y_c} |w_y|^2 \,\mathrm{d}y \le C\int_0^{y_c} (|w|^2 + |\text{source}|^2)\,\mathrm{d}y.$$

The drift floor term is $-m|w(v,0)|^2/2$, with favorable sign. Weak Hilbert limits, the variational equation, and the zero-data energy inequality give existence and uniqueness, with continuous L^2 time representative. Testing with w_v after a positive voltage interval bounds w_v, w_{yy} in L^2; repeated differentiated estimates give the local boundary regularity needed for Itô. The absence of a diffusion floor term in the variational identity encodes its Neumann trace. For continuous initial data, approximation and the maximum bound give the same solution. The full Galerkin passage and positive-time arguments are detailed for the analogous Dirichlet clock in Appendix B; here the variational space imposes only the upper-side zero value.

Until this rectangle is exited, physical voltage decreases strictly. In its inverse clock the conductance generator is $q^{-1}[a\partial_{yy} + (m-y)\partial_y]$, its discount is λ/q, and its source is r/q. Itô applied to the constructed solution cancels the drift and the Neumann derivative cancels the regulator. Physical exit time is bounded by $(v_+ - v_-)/q_-$. Stopped dynamic programming has the same boundary data, so this solution equals u. This proves the actual floor derivative for $v > 0$, without applying an interior theorem at a boundary.

A.3. **Growth for a constant jump and a bounded-source energy.** We first supply the growth assertion used in Proposition 3.4. Suppose the reset jump is A. Choose $p > \max\{1, r_0\}$ and a fixed $y_c > m, 1$ so large that

$$(\mathcal{G}_y + r_0)y^p \le -d(y + r_0 L), \qquad \mathcal{G}_y = (m-y)\partial_y + a\partial_{yy}, \quad y \ge y_c.$$

This follows from $\mathcal{G}_y y^p = -py^p + pmy^{p-1} + ap(p-1)y^{p-2}$. For $S = \inf\{t : Y_t \le y_c\}$, stopped Itô gives

$$\mathbb{E}e^{r_0 S} + \mathbb{E}\int_0^S e^{r_0 t}(Y_t + r_0 L)\,\mathrm{d}t \le C(1 + y^p).$$

Recall $X = \log[2/(2-V)] \in [0, L]$, $L = \log 2$. Between hits $\dot{X} = Y - (e^X - 1)$, and each reset decreases X by L. Including the reset jumps in $e^{r_0 t}X_t$ gives

$$L\sum_{t_j \le S} e^{r_0 t_j} \le X_0 + \int_0^S e^{r_0 t}(Y_t + r_0 L)\,\mathrm{d}t.$$

Patching the killed martingales until S consequently yields

$$u(z) = \mathbb{E}_z[e^{-\lambda S}u(Z_S)] + A\mathbb{E}_z\sum_{t_j \le S} e^{-\lambda t_j}.$$

To remove upper-conductance and time localizations before asserting this equality, choose $\beta < \beta' < 1/2$ and enlarge y_c so that $(\mathcal{G}_y + r_0 + d_0)w_{\beta'} \le 0$ above it. The upper-exit term is bounded by a fixed starting-state constant times $w_\beta((H-m)/\varepsilon)/w_{\beta'}((H-m)/\varepsilon)$, which tends to zero. The time endpoint is bounded by $e^{-d_0 t}w_{\beta'}(\eta_0)$. The displayed count bound controls the spike sum independently. Since u is bounded on $y = y_c$, this proves

$$|u(v,y)| \le C_{a,m,\lambda}(\|f\|_\beta + |A|)(1 + y^p). \tag{A.6}$$

A bounded interior source adds a controlled $\mathbb{E}\int_0^S e^{r_0 t}|r|\,\mathrm{d}t$ and gives the same conclusion. These are fixed-a growth constants.

The independent dual estimate of Section 2 needs only bounded full-process source values. We state it separately to keep its proof independent of killed-value gradients.

Proposition A.1 (Bounded-source backward energy)**.** *For bounded compact smooth r and $\lambda = s + i\theta$ with $s > 0$, let*

$$u(z) = \mathbb{E}_z \int_0^\infty e^{-\lambda t} r(Z_t)\,\mathrm{d}t.$$

It is bounded by $\|r\|_\infty / s$, has zero reset jump, satisfies (A.3)*, and has the local Neumann floor constructed above. For s larger than a fixed constant,*

$$(s - C)\,\|u\|^2_{\mathsf{H}_{-\sigma}} + a\,\|u_y\|^2_{\mathsf{H}_{-\sigma}} \le \|r\|_{\mathsf{H}_{-\sigma}}\,\|u\|_{\mathsf{H}_{-\sigma}}\,, \tag{A.7}$$

uniformly in θ. For s also larger than the forward growth bound, this value is the transpose of the full forward resolvent.

Proof. The local construction above applies to this bounded value, including its homogeneous reset and actual floor derivative. Let $\rho = e^{-2\sigma y}$. On an exhausted domain the real energy identity has volume coefficient

$$C_\sigma(y) = (\tfrac{1}{2} - \sigma)y + 1 + \sigma m + 2a\sigma^2,$$

gradient term $-a\int |u_y|^2\rho$, floor term $-(m/2 + a\sigma)\int |u(v,0)|^2\,\mathrm{d}v$, and voltage term $\frac{1}{2}\int \rho[J_F|u_F|^2 - J_0|u_0|^2]\,\mathrm{d}y$. The floor and low-conductance voltage terms are nonpositive. At high conductance $u_F = u_0$ and $J_F < J_0$, so that term is nonpositive too. Since $\sigma > 1$, C_σ is bounded above.

Here is a localization that proves the first global gradient bound rather than assuming it. Remove rectangles of voltage width δ^3 and conductance width δ around $(1,1)$ and $(0,0)$. Choose cutoffs χ_δ with $|\partial_v\chi_\delta| \le C\delta^{-3}$, $|\partial_y\chi_\delta| \le C\delta^{-1}$ and zero floor derivative. Add a large-y cutoff ζ_R and test with $\chi_\delta^2\zeta_R^2\overline{u}\rho$. Near a corner u is bounded, $|J| = O(\delta)$, and area is $O(\delta^4)$. The voltage-cutoff error is $O(\delta^2)$, the squared conductance-cutoff error is $O(\delta^2)$, and the drift-cutoff error is $O(\delta^3)$. Young's inequality absorbs the unknown gradient cross terms before any global gradient integrability is assumed. A reset-cutoff mismatch is confined to an outgoing interval with $\int J_F\,\mathrm{d}y = O(\delta^2)$. At large y, boundedness of u and the exponential weight remove all tail errors after the same absorption.

The resulting gradient bounds are uniform in δ, R, so Fatou gives $u_y \in \mathsf{H}_{-\sigma}$. Repeat the passage with this known gradient: the first-derivative cutoff errors now tend to zero by Cauchy–Schwarz, since $\|(\chi_\delta)_y u\|_2 = O(\delta)$. The exact energy inequality and the preceding signs yield (A.7). Finally Proposition 3.1 and Fubini identify the value paired against any positive initial density with the full forward Laplace resolvent. Splitting signed data and density in the weighted spaces give the claimed transpose identification. □

Appendix B. The bounded clock comparison

We prove (4.7) with no regularity modulus assumed for the terminal datum. This is the local estimate that permits the operator norm, rather than strong-operator, limit in Proposition 4.1.

Return to the scaled conductance $\eta = (y - m)/\sqrt{a}$ and the OU expression $\mathcal{A} = \partial_{\eta\eta} - \eta\partial_\eta$. The deterministic phase interval is $[0, \tau]$, with $\tau = s(\log 2)$ and scalar coefficient $Z(s) = [m - b(x(s))]^{-1}$ from (4.1). These phase objects determine the reversed coefficient $Z(\tau - t)$ used below.

Let $I_R = (-R, R)$, $\mathrm{d}\gamma = e^{-\eta^2/2}\,\mathrm{d}\eta$, $E = L^2(I_R, \mathrm{d}\gamma)$, and $V = H^1_0(I_R)$. For a smooth positive coefficient q with fixed upper and lower bounds consider

$$qu_t = (\mathcal{A} - \lambda)u, \qquad u(t, \pm R) = 0, \qquad u(0) = f. \tag{B.1}$$

The coefficient and its needed derivatives are bounded on the fixed rectangle; λ ranges in a compact set.

B.1. **Construction and positive-time estimates.** Choose a smooth Dirichlet basis with finite spans dense in V and E, and project the initial datum in E. On each finite span the q-weighted mass matrix is uniformly positive, so the finite linear system has a solution. Testing with u_N and using $\langle \mathcal{A}u, v\rangle_\gamma = -\langle u_\eta, v_\eta\rangle_\gamma$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int q|u_N|^2\,\mathrm{d}\gamma = -2\,\|(u_N)_\eta\|_2^2 - 2\,\mathrm{Re}\,\lambda\,\|u_N\|_2^2 + \int q_t|u_N|^2\,\mathrm{d}\gamma.$$

Consequently

$$\sup_{0\le t\le\tau}\|u_N(t)\|_2^2 + \int_0^\tau \|(u_N)_\eta\|_2^2\,\mathrm{d}t \le C\,\|f\|_2^2\,. \tag{B.2}$$

Take a weak subsequence in the Hilbert space $L^2_t V$. The $L^\infty_t E$ bound survives: testing the lower norm bound on every measurable time set A gives $\int_A \|u\|_E^2\,\mathrm{d}t \le C\,\|f\|_2^2\,|A|$. Integrate the finite equations by parts in time against smooth tests with values in any fixed finite span. Every term is linear, so weak convergence passes it, including q_t and the initial datum. Density gives the full variational equation. Strong time compactness is unnecessary.

Multiplication by $q^{\pm1}$ preserves V, and the equation gives $u_t \in L^2_t V'$. Time Steklov averages justify the energy identity: on a smaller interval the averages lie in $H^1_t V$, so differentiating their squared E norm is legitimate. The supremum norm of the difference of two averages is bounded by its time-average E norm squared plus twice its $L^2_t V$ norm times its derivative's $L^2_t V'$ norm. Translation continuity makes both terms tend to zero. Forward averages near the initial endpoint and backward averages near the final endpoint give a $C_t E$ representative and the limiting energy identity. Equivalently one may apply the same argument to the q-weighted norm, retaining its bounded q_t term. The initial weak datum and the energy limsup give $u(0) = f$. The zero-data energy inequality proves uniqueness. The entire Galerkin sequence therefore has this limit.

Testing the finite equation by $(u_N)_t$ instead gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\|(u_N)_\eta\|_2^2 + c\,\|(u_N)_t\|_2^2 \le C\,\|u_N\|_2^2\,.$$

Choose an earlier time with bounded gradient using (B.2), then integrate. Passing weakly gives, for each $t_* > 0$,

$$\sup_{t_*\le t\le\tau}\|u(t)\|_V^2 + \int_{t_*}^\tau \|u_t\|_2^2\,\mathrm{d}t \le C_{t_*}\,\|f\|_2^2\,. \tag{B.3}$$

The equation gives $u_{\eta\eta} \in L^2$ there. Repeated differentiated equations, with the smooth q commutators, and the one-dimensional equation give the local $C^1_t C^2_\eta$ representative on positive-time closed subrectangles. The V bound extends to each positive time by weak V compactness and strong E continuity; hence the spatial output is continuous. No compatibility at the initial Dirichlet corners is used.

For $q = 1$ the equation is autonomous. Its derivative $v = u_t$ satisfies the same homogeneous equation. Choose a time strictly before $\tau/2$ with controlled $\|v\|_2$ using (B.3), then apply the energy and positive-time estimates to v. The one-dimensional Dirichlet inequality now gives

$$\sup_{\tau/2\le t\le\tau}\|(\mathcal{A}-\lambda)u_0(t)\|_\infty \le C\,\|f\|_2\,. \tag{B.4}$$

The maximum comparison is asserted for the weak limit, not for pointwise values of a Galerkin approximation. Regularize the modulus by $\sqrt{|u|^2+\delta^2}$ and test the positive part above a scalar exponential barrier. The extra second-derivative term has nonnegative sign and can be dropped. The V energy permits the limit $\delta \downarrow 0$. This proves the maximum bound and its version with an inhomogeneous bounded source.

B.2. **Perturbation in supremum norm.** Take $q_\varepsilon = 1 + \varepsilon\eta\chi(t)$, where $\chi(t) = Z(\tau - t)$ and $\varepsilon R \|\chi\|_\infty \le 1/2$. Put $w = u_\varepsilon - u_0$ and $h = 1 - q_\varepsilon$. Its exact equation is

$$q_\varepsilon w_t = (\mathcal{A} - \lambda)w + h(\mathcal{A} - \lambda)u_0, \qquad w(0) = 0.$$

The forcing integration by parts reads

$$\langle h\mathcal{A}u_0, w\rangle_\gamma = -\int (h\overline{w_\eta} + h_\eta\overline{w})(u_0)_\eta \,\mathrm{d}\gamma.$$

Both h, h_η are $O(\varepsilon)$ on the fixed rectangle. Young's inequality and (B.2) give

$$\sup_{t\le\tau} \|w(t)\|_2^2 + \int_0^\tau \|w_\eta\|_2^2 \,\mathrm{d}t \le C\varepsilon^2 \|f\|_2^2. \tag{B.5}$$

Choose $t_1 \in [\tau/2, 3\tau/4]$ with $\|w_\eta(t_1)\|_2 \le C\varepsilon \|f\|_2$. The Dirichlet inequality on the bounded interval implies $\|w(t_1)\|_\infty \le C_R\varepsilon \|f\|_2$. On the remaining interval the divided equation has source

$$F = q_\varepsilon^{-1} h(\mathcal{A} - \lambda)u_0, \qquad \|F\|_\infty \le C\varepsilon \|f\|_2$$

by (B.4). Modulus comparison with $M' = C_\Lambda M + \|F\|_\infty$ therefore gives

$$\|w(\tau)\|_\infty \le e^{C_\Lambda \tau} \|w(t_1)\|_\infty + \int_{t_1}^\tau e^{C_\Lambda(\tau - t)} \|F(t)\|_\infty \,\mathrm{d}t \le C\varepsilon \|f\|_2 \le C_R\varepsilon \|f\|_\infty.$$

This is (4.7). Rescaling the compact family of intervals $\tau(m)$ to a unit interval makes its constants uniform for $m \in I$.

B.3. **Identification with the stopped return.** On the bounded conductance interval the physical phase speed is positive. Reverse the remaining phase time as in (4.6). The inverse-clock conductance has generator $q_\varepsilon^{-1}\mathcal{A}$ and killing rate λ/q_ε. Apply stopped Itô to the constructed positive-time classical representative. Its equation cancels the drift. The clock bounds make physical duration at most τ/q_-, and the maximum bound controls the limiting stopped values.

There is no simultaneous terminal-phase and side exit. To verify this, extend q_ε smoothly to a globally positive bounded coefficient. The unrestricted inverse-clock OU law satisfies its smooth uniformly parabolic forward equation. Interior regularity followed by narrow continuity gives a density at each positive phase time. A side exit exactly at the fixed terminal phase would put this conductance at R or $-R$, an event of probability zero. Thus continuous f is attained on surviving terminal paths and the zero Dirichlet value on earlier side exits. Dominated convergence in the stopped identity identifies the weak evolution with the killed return kernel. Approximation in E and the maximum bound give the same identification for arbitrary continuous initial data used in the comparison.

For the floor clock (A.5), replace V by $\{w \in H^1(0, y_c) : w(y_c) = 0\}$ and use Lebesgue measure. The q-weighted mass remains positive, and the first energy has the favorable floor drift $-m|w(0)|^2/2$. In the derivative estimate bound the drift term against w_v by Young's inequality before any spatial integration by parts. The same weak passage, uniqueness, and positive-time argument apply. Reflection contributes $u_y \,\mathrm{d}K = 0$ in Itô. The corresponding nondegenerate auxiliary clock law also excludes a tie of the absorbing upper side and initial voltage. This completes the construction used for the reflecting boundary in Appendix A.

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